

Language Models are Unsupervised Multitask Learners

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Abstract

Natural language processing tasks, such as question answering, machine translation, reading comprehension, and summarization, are typically approached with supervised learning on task-specific datasets. We demonstrate that language models begin to learn these tasks without any explicit supervision when trained on a new dataset of millions of webpages called WebText. When conditioned on a document plus questions, the answers generated by the language model reach 55 F1 on the CoQA dataset - matching or exceeding the performance of 3 out of 4 baseline systems without using the 127,000+ training examples. The capacity of the language model is essential to the success of zero-shot task transfer and increasing it improves performance in a log-linear fashion across tasks. Our largest model, GPT-2, is a 1.5B parameter Transformer that achieves state of the art results on 7 out of 8 tested language modeling datasets in a zero-shot setting but still underfits WebText. Samples from the model reflect these improvements and contain coherent paragraphs of text. These findings suggest a promising path towards building language processing systems which learn to perform tasks from their naturally occurring demonstrations.

1. Introduction

Machine learning systems now excel (in expectation) at tasks they are trained for by using a combination of large datasets, high-capacity models, and supervised learning (Krizhevsky et al., 2012) (Sutskever et al., 2014) (Amodei et al., 2016). Yet these systems are brittle and sensitive to slight changes in the data distribution (Recht et al., 2018) and task specification (Kirkpatrick et al., 2017). Current systems are better characterized as narrow experts rather than

competent generalists. We would like to move towards more general systems which can perform many tasks – eventually without the need to manually create and label a training dataset for each one.

The dominant approach to creating ML systems is to collect a dataset of training examples demonstrating correct behavior for a desired task, train a system to imitate these behaviors, and then test its performance on independent and identically distributed (IID) held-out examples. This has served well to make progress on narrow experts. But the often erratic behavior of captioning models (Lake et al., 2017), reading comprehension systems (Jia & Liang, 2017), and image classifiers (Alcorn et al., 2018) on the diversity and variety of possible inputs highlights some of the shortcomings of this approach.

Our suspicion is that the prevalence of single task training on single domain datasets is a major contributor to the lack of generalization observed in current systems. Progress towards robust systems with current architectures is likely to require training and measuring performance on a wide range of domains and tasks. Recently, several benchmarks have been proposed such as GLUE (Wang et al., 2018) and decaNLP (McCann et al., 2018) to begin studying this.

Multitask learning (Caruana, 1997) is a promising framework for improving general performance. However, multitask training in NLP is still nascent. Recent work reports modest performance improvements (Yogatama et al., 2019) and the two most ambitious efforts to date have trained on a total of 10 and 17 (dataset, objective) pairs respectively (McCann et al., 2018) (Bowman et al., 2018). From a meta-learning perspective, each (dataset, objective) pair is a single training example sampled from the distribution of datasets and objectives. Current ML systems need hundreds to thousands of examples to induce functions which generalize well. This suggests that multitask training many need just as many effective training pairs to realize its promise with current approaches. It will be very difficult to continue to scale the creation of datasets and the design of objectives to the degree that may be required to brute force our way there with current techniques. This motivates exploring additional setups for performing multitask learning.

The current best performing systems on language tasks

语言模型是无监督多任务学习者

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摘要

自然语言处理任务，如问答、机器翻译、阅读理解与摘要生成，通常通过在特定任务的数据集上进行监督学习来实现。我们证明，当在一个名为WebText的数百万网页新数据集上训练时，语言模型开始在没有明确监督的情况下学习这些任务。在给定文档和问题的条件下，语言模型生成的答案在CoQA数据集上达到了55 F1分数——在不使用127,000+个训练样本的情况下，匹配或超过了四分之三基线系统的性能。语言模型的能力对于零样本任务迁移的成功至关重要，增加其规模能以对数线性方式跨任务提升性能。我们最大的模型GPT-2是一个拥有15亿参数的Transformer，在零样本设置下，它在8个测试语言建模数据集集中的7个上取得了最先进的结果，但仍未充分拟合WebText。模型生成的样本反映了这些改进，并包含连贯的文本段落。这些发现为构建能够从自然发生的演示中学习执行任务的语言处理系统指明了一条充满希望的道路。

合格的通才。我们希望朝着更通用的系统发展，这些系统能够执行多种任务——最终无需为每项任务手动创建和标注训练数据集。

创建机器学习系统的主流方法是收集一个训练示例数据集，展示期望任务中的正确行为，训练系统模仿这些行为，然后在独立同分布（IID）的保留示例上测试其性能。这种方法在推动专业领域进展方面表现良好。但字幕模型（Lake等人，2017）、阅读理解系统（Jia与Liang，2017）和图像分类器（Alcorn等人，2018）在面对多样化和各种可能输入时经常表现不稳定，突显了这种方法的某些不足。

我们怀疑，当前系统泛化能力不足的主要原因在于普遍采用单领域数据集进行单任务训练。要基于现有架构实现稳健系统，很可能需要在广泛领域和任务上进行训练和性能评估。最近，GLUE（Wang et al., 2018）和decaNLP（McCann et., 2018）等基准测试的提出，已为相关研究拉开了序幕。

多任务学习（Caruana, 1997）是一个提升整体性能的有前景的框架。然而，自然语言处理中的多任务训练仍处于起步阶段。近期研究仅报告了有限的性能提升（Yogatama等人，2019），而迄今为止两项最雄心勃勃的尝试分别仅使用了总计10对和17对（数据集，目标）进行训练（McCann等人，2018）（Bowman等人，2018）。从元学习的角度来看，每个（数据集，目标）对都是从数据集和目标分布中采样的单个训练样本。当前的机器学习系统需要数百到数千个样本才能归纳出泛化良好的函数。这表明，多任务训练可能需要同样多的有效训练对，才能在当前方法下实现其潜力。若继续按当前技术路径，通过简单堆叠数据集创建和目标设计来达到所需规模，将极为困难。这促使我们探索更多进行多任务学习的实验设置。

当前在语言任务上表现最佳的系统

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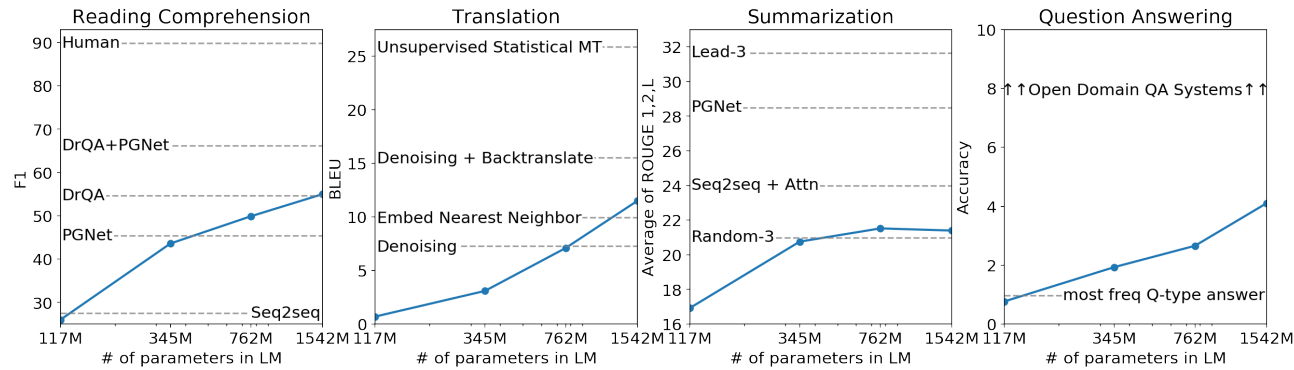


Figure 1. Zero-shot task performance of WebText LMs as a function of model size on many NLP tasks. Reading Comprehension results are on CoQA (Reddy et al., 2018), translation on WMT-14 Fr-En (Artetxe et al., 2017), summarization on CNN and Daily Mail (See et al., 2017), and Question Answering on Natural Questions (Kwiatkowski et al., 2019). Section 3 contains detailed descriptions of each result.

utilize a combination of pre-training and supervised fine-tuning. This approach has a long history with a trend towards more flexible forms of transfer. First, word vectors were learned and used as inputs to task-specific architectures (Mikolov et al., 2013) (Collobert et al., 2011), then the contextual representations of recurrent networks were transferred (Dai & Le, 2015) (Peters et al., 2018), and recent work suggests that task-specific architectures are no longer necessary and transferring many self-attention blocks is sufficient (Radford et al., 2018) (Devlin et al., 2018).

These methods still require supervised training in order to perform a task. When only minimal or no supervised data is available, another line of work has demonstrated the promise of language models to perform specific tasks, such as commonsense reasoning (Schwartz et al., 2017) and sentiment analysis (Radford et al., 2017).

In this paper, we connect these two lines of work and continue the trend of more general methods of transfer. We demonstrate language models can perform down-stream tasks in a zero-shot setting – without any parameter or architecture modification. We demonstrate this approach shows potential by highlighting the ability of language models to perform a wide range of tasks in a zero-shot setting. We achieve promising, competitive, and state of the art results depending on the task.

2. Approach

At the core of our approach is language modeling. Language modeling is usually framed as unsupervised distribution estimation from a set of examples $(x_1, x_2, \dots, x_n)$ each composed of variable length sequences of symbols $(s_1, s_2, \dots, s_n)$. Since language has a natural sequential ordering, it is common to factorize the joint probabilities over

symbols as the product of conditional probabilities (Jelinek & Mercer, 1980) (Bengio et al., 2003):

$$p(x) = \prod_{i=1}^n p(s_i | s_1, \dots, s_{i-1}) \quad (1)$$

This approach allows for tractable sampling from and estimation of $p(x)$ as well as any conditionals of the form $p(s_{n-k}, \dots, s_n | s_1, \dots, s_{n-k-1})$. In recent years, there have been significant improvements in the expressiveness of models that can compute these conditional probabilities, such as self-attention architectures like the Transformer (Vaswani et al., 2017).

Learning to perform a single task can be expressed in a probabilistic framework as estimating a conditional distribution $p(output | input)$. Since a general system should be able to perform many different tasks, even for the same input, it should condition not only on the input but also on the task to be performed. That is, it should model $p(output | input, task)$. This has been variously formalized in multitask and meta-learning settings. Task conditioning is often implemented at an architectural level, such as the task specific encoders and decoders in (Kaiser et al., 2017) or at an algorithmic level such as the inner and outer loop optimization framework of MAML (Finn et al., 2017). But as exemplified in McCann et al. (2018), language provides a flexible way to specify tasks, inputs, and outputs all as a sequence of symbols. For example, a translation training example can be written as the sequence (translate to french, english text, french text). Likewise, a reading comprehension training example can be written as (answer the question, document, question, answer). McCann et al. (2018) demonstrated it was possible to train a single model, the MQAN,

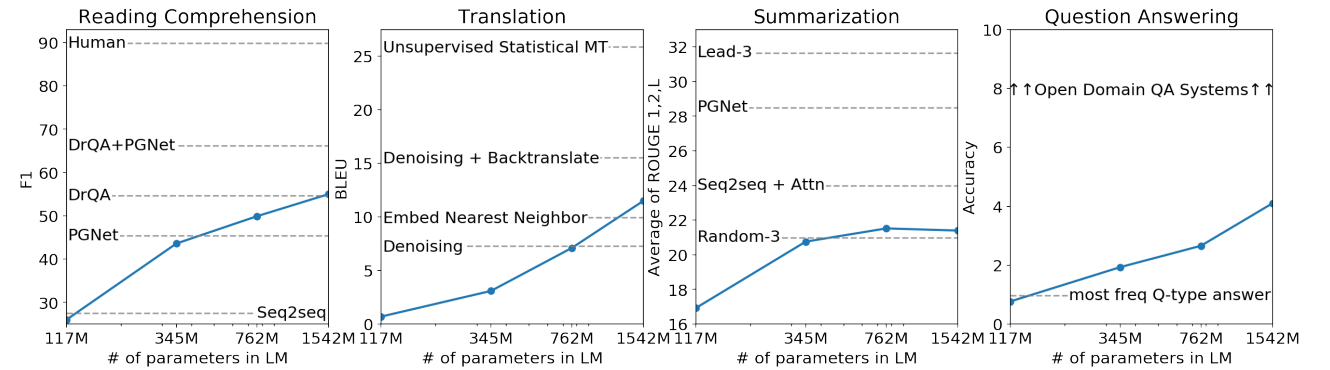


图1. WebText语言模型在多种NLP任务上的零样本任务性能随模型规模的变化情况。阅读理解结果基于CoQA数据集（Reddy等人，2018），翻译结果基于WMT-14法英数据集（Artetxe等人，2017），摘要生成结果基于CNN和《每日邮报》数据集（See等人，2017），问答结果基于自然问题数据集（Kwiatkowski等人，2019）。第3节包含对各结果的详细说明。

采用预训练与监督微调相结合的方法。这一方法历史悠久，其趋势是向更灵活的迁移形式发展。最初是学习词向量并将其作为任务特定架构的输入（Mikolov等人，2013年）（Collobert等人，2011年），随后迁移了循环网络的上下文表示（Dai & Le, 2015年）（Peters等人，2018年），而近期研究表明，任务特定架构已不再必要，迁移多个自注意力块便已足够（Radford等人，2018年）（Devlin等人，2018年）。

这些方法仍需要监督训练来执行任务。当仅有少量或没有监督数据可用时，另一系列研究展示了语言模型在执行特定任务（如常识推理（Schwartz等人，2017）和情感分析（Radford等人，2017））方面的潜力。

在本文中，我们将这两条研究脉络联系起来，并延续了更通用迁移方法的发展趋势。我们证明了语言模型能够在零样本设置下执行下游任务——无需任何参数或架构调整。通过强调语言模型在零样本设置下执行广泛任务的能力，我们证明了这种方法具有潜力。根据任务的不同，我们取得了有前景、具有竞争力以及最先进的结果。

2. 方法

我们方法的核心是语言建模。语言建模通常被定义为从一组示例 $(x_1, x_2, \dots, x_n)$ 中进行无监督分布估计，每个示例由可变长度的符号序列 $(s_1, s_2, \dots, s_n)$ 组成。由于语言具有自然的顺序结构，通常会对联合概率进行分解。

符号作为条件概率的乘积（Jelinek & Mercer, 1980）（Bengio et al., 2003）：

$$p(x) = \prod_{i=1}^n p(s_i | s_1, \dots, s_{i-1}) \quad (1)$$

这种方法允许从 $p(x)$ 以及任何形式为 $p(s_{n-k}, \dots, s_n | s_1, \dots, s_{n-k-1})$ 的条件分布中进行可处理的采样和估计。近年来，能够计算这些条件概率的模型表达能力有了显著提升，例如Transformer等自注意力架构（Vaswani等人，2017年）。

学习执行单一任务可以在概率框架中表达为估计条件分布 $p(output | input)$ 。由于一个通用系统应能执行许多不同的任务，即使对于相同的输入，它不仅应基于输入进行条件化，还应基于要执行的任务。也就是说，它应建模 $p(output | input, task)$ 。这在多任务和元学习设置中已有多种形式化实现。任务条件化通常在架构层面实现，例如（Kaiser等人，2017）中的任务特定编码器和解码器，或在算法层面实现，例如MAML（Finn等人，2017）的内外循环优化框架。但如McCann等人（2018）所示，语言提供了一种灵活的方式，将任务、输入和输出全部指定为符号序列。例如，一个翻译训练示例可以写为序列（翻译成法语，英语文本，法语文本）。同样，一个阅读理解训练示例可以写为（回答问题，文档，问题，答案）。McCann等人（2018）证明了训练单一模型MQAN是可能的，

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to infer and perform many different tasks on examples with this type of format. Language modeling is also able to, in principle, learn the tasks of McCann et al. (2018) without the need for explicit supervision of which symbols are the outputs to be predicted. Since the supervised objective is the the same as the unsupervised objective but only evaluated on a subset of the sequence, the global minimum of the unsupervised objective is also the global minimum of the supervised objective. In this slightly toy setting, the concerns with density estimation as a principled training objective discussed in (Sutskever et al., 2015) are side stepped. The problem instead becomes whether we are able to, in practice, optimize the unsupervised objective to convergence. Preliminary experiments confirmed that sufficiently large language models are able to perform multitask learning in this toy-ish setup but learning is much slower than in explicitly supervised approaches. While it is a large step from the well-posed setup described above to the messiness of “language in the wild”, Weston (2016) argues, in the context of dialog, for the need to develop systems capable of learning from natural language directly and demonstrated a proof of concept – learning a QA task without a reward signal by using forward prediction of a teacher’s outputs. While dialog is an attractive approach, we worry it is overly restrictive. The internet contains a vast amount of information that is passively available without the need for interactive communication. Our speculation is that a language model with sufficient capacity will begin to learn to infer and perform the tasks demonstrated in natural language sequences in order to better predict them, regardless of their method of procurement. If a language model is able to do this it will be, in effect, performing unsupervised multitask learning. We test whether this is the case by analyzing the performance of language models in a zero-shot setting on a wide variety of tasks.
”I’m not the cleverest man in the world, but like they say in French: Je ne suis pas un imbécile [I’m not a fool]. In a now-deleted post from Aug. 16, Soheil Eid, Tory candidate in the riding of Joliette, wrote in French: ”Mentez mentez, il en restera toujours quelque chose,” which translates as, ”Lie lie and something will always remain.” “I hate the word ‘perfume,’” Burr says. ‘It’s somewhat better in French: ‘parfum.’ If listened carefully at 29:55, a conversation can be heard between two guys in French: “-Comment on fait pour aller de l’autre coté? -Quel autre coté?”, which means “- How do you get to the other side? - What side?”. If this sounds like a bit of a stretch, consider this question in French: As-tu aller au cinéma?, or Did you go to the movies?, which literally translates as Have-you to go to movies/theater? “Brevet Sans Garantie Du Gouvernement”, translated to English: “Patented without government warranty”.

Table 1. Examples of naturally occurring demonstrations of English to French and French to English translation found throughout the WebText training set.

Common Crawl. [Trinh & Le \(2018\)](#)’s best results were achieved using a small subsample of Common Crawl which included only documents most similar to their target dataset, the Winograd Schema Challenge. While this is a pragmatic approach to improve performance on a specific task, we want to avoid making assumptions about the tasks to be performed ahead of time.

Instead, we created a new web scrape which emphasizes document quality. To do this we only scraped web pages which have been curated/filtered by humans. Manually filtering a full web scrape would be exceptionally expensive so as a starting point, we scraped all outbound links from Reddit, a social media platform, which received at least 3 karma. This can be thought of as a heuristic indicator for whether other users found the link interesting, educational, or just funny.

The resulting dataset, WebText, contains the text subset of these 45 million links. To extract the text from HTML responses we use a combination of the Dragnet ([Peters & Lecocq, 2013](#)) and Newspaper¹ content extractors. All results presented in this paper use a preliminary version of WebText which does not include links created after Dec 2017 and which after de-duplication and some heuristic based cleaning contains slightly over 8 million documents for a total of 40 GB of text. We removed all Wikipedia documents from WebText since it is a common data source for other datasets and could complicate analysis due to over-

¹<https://github.com/codelucas/newspaper>

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推断并在此类格式的示例上执行多种不同的任务。 语言建模原则上也能够学习McCann等人（2018）的任务，而无需明确监督哪些符号是需要预测的输出。由于监督目标与非监督目标相同，但仅在序列的子集上进行评估，因此非监督目标的全局最小值也是监督目标的全局最小值。在这个略显简化的设定中，（Sutskever等人，2015）讨论的将密度估计作为原则性训练目标的问题被规避了。问题反而变成了我们是否能够在实践中优化非监督目标直至收敛。初步实验证实，足够大的语言模型能够在这种简化设置中进行多任务学习，但学习速度远慢于显式监督方法。 尽管从上述定义明确的情境到"野外语言"的混乱状态是一个巨大的跨越，但韦斯顿（2016）在对话语境中提出，有必要开发能够直接从自然语言中学习的系统，并通过预测教师输出的前向预测方法，展示了无需奖励信号即可完成问答任务的概念验证。虽然对话是一种颇具吸引力的途径，但我们担心其限制性过强。互联网蕴含着海量被动可获取的信息，无需交互式沟通即可获得。我们推测，具备足够能力的语言模型将开始学习推断并执行自然语言序列中展示的任务，以更好地预测这些序列，无论其获取方式如何。如果语言模型能够做到这一点，实际上就是在进行无监督多任务学习。我们通过分析语言模型在零样本设置下广泛任务中的表现，来验证这一假设是否成立。 2.1. 训练数据集 先前的大多数工作都是在单一文本领域训练语言模型，例如新闻文章（Jozefowicz等人，2016）、维基百科（Merity等人，2016）或小说书籍（Kiros等人，2015）。我们的方法则鼓励构建尽可能庞大且多样化的数据集，以便在尽可能多的领域和情境中收集任务的自然语言示例。 一个前景广阔、多样且几乎无限的文本来源是网络抓取数据，例如Common Crawl。尽管这些存档的规模比当前的语言建模数据集大几个数量级，但它们存在显著的数据质量问题。Trinh & Le（2018）在他们关于常识推理的研究中使用了Common Crawl，但指出大量文档“内容大多难以理解”。我们在初步实验中也观察到了类似的数据问题。
“我不是世界上最聪明的人，但正如法语里说的：Je ne suis pas un imbécile [我不是傻瓜]。 在8月16日一篇现已删除的帖子中，若列特选区的保守党候选人索海尔·艾德用法语写道：“Mentez mentez, il en restera toujours quelque chose", 意为"谎言重复千遍，总有痕迹残留"。 “我讨厌‘香水’这个词，”伯尔说。“法语里稍微好一点：‘parfum’。” 如果仔细听29分55秒处，可以听到两个家伙用法语交谈： “-Comment on fait pour aller de l’ autre coté? -Quel autre coté?”, 意思是 “-怎么去另一边? -哪一边?” 。 如果这听起来有点牵强，不妨想想这个法语问题：As-tu aller au cinéma?, 或者 “Did you go to the movies?”, 其字面翻译是 “你-有 去 电影/剧院吗?” “Brevet Sans Garantie Du Gouvernement”，英文译为：“Patented without government warranty”。

表1. 在WebText训练集中发现的英语到法语和法语到英语翻译的自然演示示例。

Common Crawl. Trinh & Le（2018）的最佳结果是通过使用Common Crawl的一个小子样本实现的，该子样本仅包含与他们的目标数据集——Winograd Schema Challenge最相似的文档。虽然这是一种提高特定任务性能的实用方法，但我们希望避免提前对要执行的任务做出假设。

相反，我们创建了一个新的网络抓取系统，强调文档质量。为此，我们只抓取经过人工筛选/整理的网页。手动筛选整个网络抓取内容将异常昂贵，因此作为起点，我们抓取了社交媒体平台Reddit上所有获得至少3点“karma”的外链。这可以被视为一种启发式指标，用于判断其他用户是否认为该链接有趣、有教育意义或只是好笑。

由此得到的数据集WebText包含了这4500万个链接的文本子集。为了从HTML响应中提取文本，我们结合使用了Dragnet（Peters & Lecocq, 2013）和Newspaper¹内容提取器。本文展示的所有结果均基于WebText的初步版本，该版本未包含2017年12月之后创建的链接，经过去重和基于启发式的清理后，包含略超过800万份文档，文本总量达40 GB。我们从WebText中移除了所有维基百科文档，因为这是其他数据集的常见来源，且可能因过度——

¹<https://github.com/codelucas/newspaper>

lapping training data with test evaluation tasks.

2.2. Input Representation

A general language model (LM) should be able to compute the probability of (and also generate) any string. Current large scale LMs include pre-processing steps such as lower-casing, tokenization, and out-of-vocabulary tokens which restrict the space of model-able strings. While processing Unicode strings as a sequence of UTF-8 bytes elegantly fulfills this requirement as exemplified in work such as Gillick et al. (2015), current byte-level LMs are not competitive with word-level LMs on large scale datasets such as the One Billion Word Benchmark (Al-Rfou et al., 2018). We observed a similar performance gap in our own attempts to train standard byte-level LMs on WebText.

Byte Pair Encoding (BPE) (Sennrich et al., 2015) is a practical middle ground between character and word level language modeling which effectively interpolates between word level inputs for frequent symbol sequences and character level inputs for infrequent symbol sequences. Despite its name, reference BPE implementations often operate on Unicode code points and not byte sequences. These implementations would require including the full space of Unicode symbols in order to model all Unicode strings. This would result in a base vocabulary of over 130,000 before any multi-symbol tokens are added. This is prohibitively large compared to the 32,000 to 64,000 token vocabularies often used with BPE. In contrast, a byte-level version of BPE only requires a base vocabulary of size 256. However, directly applying BPE to the byte sequence results in sub-optimal merges due to BPE using a greedy frequency based heuristic for building the token vocabulary. We observed BPE including many versions of common words like dog since they occur in many variations such as dog. dog! dog? . This results in a sub-optimal allocation of limited vocabulary slots and model capacity. To avoid this, we prevent BPE from merging across character categories for any byte sequence. We add an exception for spaces which significantly improves the compression efficiency while adding only minimal fragmentation of words across multiple vocab tokens.

This input representation allows us to combine the empirical benefits of word-level LMs with the generality of byte-level approaches. Since our approach can assign a probability to any Unicode string, this allows us to evaluate our LMs on any dataset regardless of pre-processing, tokenization, or vocab size.

2.3. Model

We use a Transformer (Vaswani et al., 2017) based architecture for our LMs. The model largely follows the details of the OpenAI GPT model (Radford et al., 2018) with a

Parameters	Layers	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542M	48	1600

Table 2. Architecture hyperparameters for the 4 model sizes.

few modifications. Layer normalization (Ba et al., 2016) was moved to the input of each sub-block, similar to a pre-activation residual network (He et al., 2016) and an additional layer normalization was added after the final self-attention block. A modified initialization which accounts for the accumulation on the residual path with model depth is used. We scale the weights of residual layers at initialization by a factor of $1/\sqrt{N}$ where N is the number of residual layers. The vocabulary is expanded to 50,257. We also increase the context size from 512 to 1024 tokens and a larger batchsize of 512 is used.

3. Experiments

We trained and benchmarked four LMs with approximately log-uniformly spaced sizes. The architectures are summarized in Table 2. The smallest model is equivalent to the original GPT, and the second smallest equivalent to the largest model from BERT (Devlin et al., 2018). Our largest model, which we call GPT-2, has over an order of magnitude more parameters than GPT. The learning rate of each model was manually tuned for the best perplexity on a 5% held-out sample of WebText. All models still underfit WebText and held-out perplexity has as of yet improved given more training time.

3.1. Language Modeling

As an initial step towards zero-shot task transfer, we are interested in understanding how WebText LM’s perform at zero-shot domain transfer on the primary task they are trained for – language modeling. Since our model operates on a byte level and does not require lossy pre-processing or tokenization, we can evaluate it on any language model benchmark. Results on language modeling datasets are commonly reported in a quantity which is a scaled or exponentiated version of the average negative log probability per canonical prediction unit - usually a character, a byte, or a word. We evaluate the same quantity by computing the log-probability of a dataset according to a WebText LM and dividing by the number of canonical units. For many of these datasets, WebText LMs would be tested significantly out-of-distribution, having to predict aggressively standardized text, tokenization artifacts such as disconnected punctuation and contractions, shuffled sentences, and even the string

在训练数据中混入测试评估任务。

2.2. 输入表示

一个通用的语言模型（LM）应该能够计算（并生成）任何字符串的概率。当前的大规模语言模型包含预处理步骤，如小写转换、分词和词汇表外标记处理，这些步骤限制了模型可处理的字符串范围。虽然将Unicode字符串作为UTF-8字节序列处理优雅地满足了这一要求，例如Gillick等人（2015年）的工作所示，但当前的字节级语言模型在大型数据集（如十亿词基准测试，Al-Rfou等人，2018年）上仍无法与词级语言模型竞争。我们在尝试训练WebText上的标准字节级语言模型时，也观察到了类似的性能差距。

字节对编码（BPE）（Sennrich等人，2015）是字符级和词级语言建模之间的一种实用折中方案，它有效地在频繁符号序列的词级输入和不频繁符号序列的字符级输入之间进行插值。尽管其名称如此，参考的BPE实现通常操作的是Unicode码点而非字节序列。这些实现需要包含完整的Unicode符号空间才能对所有Unicode字符串进行建模。这会导致在添加任何多符号标记之前，基础词汇表就超过130,000个。与BPE通常使用的32,000到64,000个标记的词汇表相比，这过于庞大。相比之下，字节级的BPE版本仅需要大小为256的基础词汇总表。然而，由于BPE使用基于贪婪频率的启发式方法来构建标记词汇表，直接将BPE应用于字节序列会导致次优的合并。我们观察到BPE包含了许多常见单词的多个版本，例如dog，因为它们以多种变体出现，如dog.、dog!、dog?。这导致有限词汇表槽位和模型容量的次优分配。为了避免这种情况，我们阻止BPE在任何字节序列中跨字符类别进行合并。我们为空格添加了一个例外，这显著提高了压缩效率，同时仅使单词在多个词汇标记中的碎片化程度最小化。

这种输入表示使我们能够将词级语言模型的实证优势与字节级方法的普适性结合起来。由于我们的方法能够为任何Unicode字符串分配概率，这使得我们可以在任何数据集上评估语言模型，无需考虑预处理、分词或词汇量大小。

2.3. 模型

我们采用基于Transformer（Vaswani等人，2017）的架构构建语言模型。该模型主要遵循OpenAI GPT模型（Radford等人，2018）的设计细节，并在此基础上

Parameters	Layers	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542M	48	1600

表2. 四种模型规模的架构超参数。

进行了少量修改。层归一化（Ba等人，2016）被移至每个子块的输入处，类似于预激活残差网络（He等人，2016），并在最终的自注意力块后增加了一个额外的层归一化。采用了一种改进的初始化方法，该方法考虑了模型深度对残差路径累积的影响。在初始化时，我们将残差层的权重按 $1/\sqrt{N}$ 的比例缩放，其中 N 是残差层的数量。词汇表扩展到50,257个词元。同时，我们将上下文大小从512个词元增加到1024个词元，并使用了更大的批次大小512。

3. 实验

我们训练并基准测试了四个大小近似对数均匀分布的语言模型。架构总结在表2中。最小的模型相当于原始的GPT，第二小的模型相当于BERT的最大模型（Devlin等人，2018）。我们最大的模型，我们称之为GPT-2，其参数数量比GPT多一个数量级以上。每个模型的学习率都经过手动调整，以在WebText的5%保留样本上获得最佳困惑度。所有模型仍然对WebText欠拟合，且随着训练时间的增加，保留样本的困惑度仍在持续改善。

3.1. 语言建模

作为迈向零样本任务迁移的第一步，我们关注于探究WebText语言模型在零样本领域迁移中的表现，尤其是在其核心训练任务——语言建模上的表现。由于我们的模型在字节级别运行，无需进行有损预处理或分词，因此可以在任何语言模型基准上进行评估。语言建模数据集的结果通常以某种量化指标呈现，该指标是每个标准预测单元（通常为字符、字节或单词）平均负对数概率的缩放或指数化版本。我们通过计算WebText语言模型对数据集的对数概率，再除以标准单元数量来评估相同指标。对于其中许多数据集，WebText语言模型将面临显著超出分布范围的测试：需要预测经过激进标准化的文本、分词衍生物（如断开的标点和缩略形式）、乱序句子，甚至字符串

Language Models are Unsupervised Multitask Learners										
	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText2 (PPL)	PTB (PPL)	enwik8 (BPB)	text8 (BPC)	WikiText103 (PPL)	1BW (PPL)
SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	1.08	18.3	21.8
117M	35.13	45.99	87.65	83.4	29.41	65.85	1.16	1.17	37.50	75.20
345M	15.60	55.48	92.35	87.1	22.76	47.33	1.01	1.06	26.37	55.72
762M	10.87	60.12	93.45	88.0	19.93	40.31	0.97	1.02	22.05	44.575
1542M	8.63	63.24	93.30	89.05	18.34	35.76	0.93	0.98	17.48	42.16

Table 3. Zero-shot results on many datasets. No training or fine-tuning was performed for any of these results. PTB and WikiText-2 results are from (Gong et al., 2018). CBT results are from (Bajgar et al., 2016). LAMBADA accuracy result is from (Hoang et al., 2018) and LAMBADA perplexity result is from (Grave et al., 2016). Other results are from (Dai et al., 2019).

<UNK> which is extremely rare in WebText - occurring only 26 times in 40 billion bytes. We report our main results in Table 3 using invertible de-tokenizers which remove as many of these tokenization / pre-processing artifacts as possible. Since these de-tokenizers are invertible, we can still calculate the log probability of a dataset and they can be thought of as a simple form of domain adaptation. We observe gains of 2.5 to 5 perplexity for GPT-2 with these de-tokenizers.

WebText LMs transfer well across domains and datasets, improving the state of the art on 7 out of the 8 datasets in a zero-shot setting. Large improvements are noticed on small datasets such as Penn Treebank and WikiText-2 which have only 1 to 2 million training tokens. Large improvements are also noticed on datasets created to measure long-term dependencies like LAMBADA (Paperno et al., 2016) and the Children’s Book Test (Hill et al., 2015). Our model is still significantly worse than prior work on the One Billion Word Benchmark (Chelba et al., 2013). This is likely due to a combination of it being both the largest dataset and having some of the most destructive pre-processing - 1BW’s sentence level shuffling removes all long-range structure.

3.2. Children’s Book Test

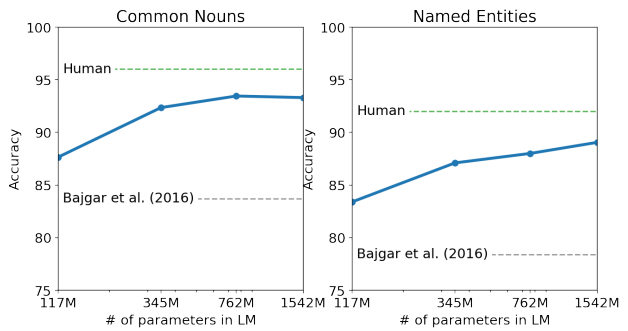


Figure 2. Performance on the Children’s Book Test as a function of model capacity. Human performance are from Bajgar et al. (2016), instead of the much lower estimates from the original paper.

The Children’s Book Test (CBT) (Hill et al., 2015) was created to examine the performance of LMs on different categories of words: named entities, nouns, verbs, and prepositions. Rather than reporting perplexity as an evaluation metric, CBT reports accuracy on an automatically constructed cloze test where the task is to predict which of 10 possible choices for an omitted word is correct. Following the LM approach introduced in the original paper, we compute the probability of each choice and the rest of the sentence conditioned on this choice according to the LM, and predict the one with the highest probability. As seen in Figure 2 performance steadily improves as model size is increased and closes the majority of the gap to human performance on this test. Data overlap analysis showed one of the CBT test set books, The Jungle Book by Rudyard Kipling, is in WebText, so we report results on the validation set which has no significant overlap. GPT-2 achieves new state of the art results of 93.3% on common nouns and 89.1% on named entities. A de-tokenizer was applied to remove PTB style tokenization artifacts from CBT.

3.3. LAMBADA

The LAMBADA dataset (Paperno et al., 2016) tests the ability of systems to model long-range dependencies in text. The task is to predict the final word of sentences which require at least 50 tokens of context for a human to successfully predict. GPT-2 improves the state of the art from 99.8 (Grave et al., 2016) to 8.6 perplexity and increases the accuracy of LMs on this test from 19% (Dehghani et al., 2018) to 52.66%. Investigating GPT-2’s errors showed most predictions are valid continuations of the sentence, but are not valid final words. This suggests that the LM is not using the additional useful constraint that the word must be the final of the sentence. Adding a stop-word filter as an approximation to this further increases accuracy to 63.24%, improving the overall state of the art on this task by 4%. The previous state of the art (Hoang et al., 2018) used a different restricted prediction setting where the outputs of the model were constrained to only words that appeared in the context. For GPT-2, this restriction is harmful rather than helpful

语言模型是无监督多任务学习者										
	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText2 (PPL)	PTB (PPL)	enwik8 (BPB)	text8 (BPC)	WikiText103 (PPL)	1BW (PPL)
SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	1.08	18.3	21.8
117M	35.13	45.99	87.65	83.4	29.41	65.85	1.16	1.17	37.50	75.20
345M	15.60	55.48	92.35	87.1	22.76	47.33	1.01	1.06	26.37	55.72
762M	10.87	60.12	93.45	88.0	19.93	40.31	0.97	1.02	22.05	44.575
1542M	8.63	63.24	93.30	89.05	18.34	35.76	0.93	0.98	17.48	42.16

表3. 多个数据集上的零样本结果。这些结果均未进行任何训练或微调。PTB和WikiText-2的结果来自(Gong等人, 2018)。CBT结果来自(Bajgar等人, 2016)。LAMBADA准确率结果来自(Hoang等人, 2018), LAMBADA困惑度结果来自(Grave等人, 2016)。其他结果来自(Dai等人, 2019)。

<UNK>在WebText中极为罕见——在400亿字节中仅出现26次。我们在表3中报告了主要结果，使用了可逆的去标记器，以尽可能消除这些标记化/预处理带来的伪影。由于这些去标记器是可逆的，我们仍然可以计算数据集的概率对数，它们可以被视为一种简单的领域适应形式。我们观察到，使用这些去标记器后，GPT-2的困惑度提升了2.5到5个点。

WebText语言模型在不同领域和数据集间展现出良好的迁移能力，在零样本设定下，其在8个数据集上的7个上实现了技术水平的提升。在训练标记仅有100万到200万的小型数据集上，如Penn Treebank和WikiText-2，观察到了显著的改进。同样，在用于衡量长期依赖关系的数据集上，如LAMBADA (Paperno等人, 2016)和儿童图书测试 (Hill等人, 2015)，也观察到了大幅度的提升。然而，在十亿词基准测试 (Chelba等人, 2013) 上，我们的模型仍显著落后于先前的研究。这可能是由于该数据集规模最大，且经过了最具破坏性的预处理——十亿词基准的句子级打乱破坏了所有的长程结构。

3.2. 儿童读物测试

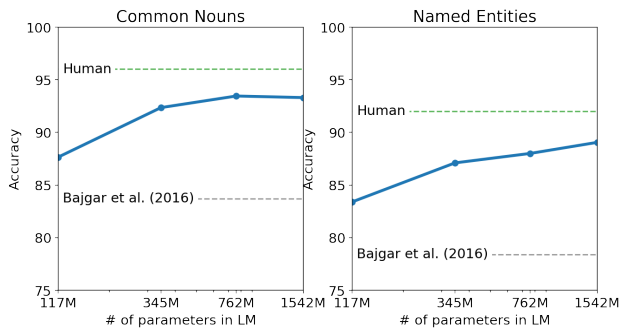


图2. 儿童图书测试表现随模型容量的变化关系。人类表现数据来自Bajgar等人 (2016) 的研究，而非原始论文中低得多的估计值。

儿童图书测试 (CBT) (Hill等人, 2015) 旨在检验语言模型在不同词类上的表现：命名实体、名词、动词和介词。CBT不采用困惑度作为评估指标，而是报告在自动构建的完形填空测试中的准确率，该测试的任务是从10个可能的选项中预测被省略单词的正确选项。遵循原论文中引入的语言模型方法，我们根据语言模型计算每个选项及句子其余部分在该选项条件下的概率，并预测概率最高的选项。如图2所示，随着模型规模的增加，性能稳步提升，并在此测试中大幅缩小了与人类表现的差距。数据重叠分析显示，CBT测试集中的一本书——鲁德亚德·吉卜林的《丛林之书》——存在于WebText中，因此我们报告在验证集上的结果，该集合没有显著重叠。GPT-2在普通名词上取得了93.3%的最新最优结果，在命名实体上取得了89.1%的成绩。应用了去标记化器以移除CBT中PTB风格的标记化伪影。

3.3. LAMBADA

LAMBADA数据集 (Paperno等人, 2016年) 用于测试系统对文本中长距离依赖关系的建模能力。该任务要求预测句子的最后一个词，这些句子至少需要50个词元的上下文，人类才能成功预测。GPT-2将困惑度指标从99.8 (Grave等人, 2016年) 提升至8.6，并将语言模型在此测试中的准确率从19% (Dehghani等人, 2018年) 提高到52.66%。对GPT-2错误的分析显示，大多数预测是句子的有效延续，但并非有效的结尾词。这表明语言模型未能充分利用“该词必须是句子结尾”这一附加约束条件。通过添加停用词过滤器作为近似处理，准确率进一步提升至63.24%，使该任务的整体技术水平提高了4%。先前的最优方法 (Hoang等人, 2018年) 采用了不同的受限预测设置，将模型输出限制为仅出现在上下文中的词汇。对于GPT-2而言，这种限制反而会产生负面影响。

since 19% of answers are not in context. We use a version of the dataset without preprocessing.

3.4. Winograd Schema Challenge

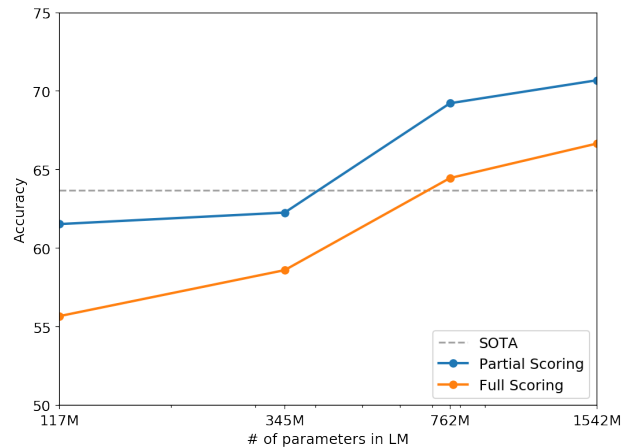


Figure 3. Performance on the Winograd Schema Challenge as a function of model capacity.

The Winograd Schema challenge (Levesque et al., 2012) was constructed to measure the capability of a system to perform commonsense reasoning by measuring its ability to resolve ambiguities in text. Recently Trinh & Le (2018) demonstrated significant progress on this challenge using LMs, by predicting the resolution of the ambiguity with higher probability. We follow their problem formulation and visualize the performance of our models with both full and partial scoring techniques in Figure 3. GPT-2 improves state of the art accuracy by 7%, achieving 70.70%. The dataset is quite small with only 273 examples so we recommend reading Trichelair et al. (2018) to help contextualize this result.

3.5. Reading Comprehension

The Conversation Question Answering dataset (CoQA) Reddy et al. (2018) consists of documents from 7 different domains paired with natural language dialogues between a question asker and a question answerer about the document. CoQA tests reading comprehension capabilities and also the ability of models to answer questions that depend on conversation history (such as “Why?”).

Greedy decoding from GPT-2 when conditioned on a document, the history of the associated conversation, and a final token A: achieves 55 F1 on the development set. This matches or exceeds the performance of 3 out of 4 baseline systems without using the 127,000+ manually collected question answer pairs those baselines were trained on. The supervised SOTA, a BERT based system (Devlin et al.,

	R-1	R-2	R-L	R-AVG
Bottom-Up Sum	41.22	18.68	38.34	32.75
Lede-3	40.38	17.66	36.62	31.55
Seq2Seq + Attn	31.33	11.81	28.83	23.99
GPT-2 TL;DR:	29.34	8.27	26.58	21.40
Random-3	28.78	8.63	25.52	20.98
GPT-2 no hint	21.58	4.03	19.47	15.03

Table 4. Summarization performance as measured by ROUGE F1 metrics on the CNN and Daily Mail dataset. Bottom-Up Sum is the SOTA model from (Gehrmann et al., 2018)

2018), is nearing the 89 F1 performance of humans. While GPT-2’s performance is exciting for a system without any supervised training, some inspection of its answers and errors suggests GPT-2 often uses simple retrieval based heuristics such as *answer with a name from the document in response to a who question*.

3.6. Summarization

We test GPT-2’s ability to perform summarization on the CNN and Daily Mail dataset (Nallapati et al., 2016). To induce summarization behavior we add the text TL;DR: after the article and generate 100 tokens with Top- k random sampling (Fan et al., 2018) with $k = 2$ which reduces repetition and encourages more abstractive summaries than greedy decoding. We use the first 3 generated sentences in these 100 tokens as the summary. While qualitatively the generations resemble summaries, as shown in Table 14, they often focus on recent content from the article or confuse specific details such as how many cars were involved in a crash or whether a logo was on a hat or shirt. On the commonly reported ROUGE 1,2,L metrics the generated summaries only begin to approach the performance of classic neural baselines and just barely outperforms selecting 3 random sentences from the article. GPT-2’s performance drops by 6.4 points on the aggregate metric when the task hint is removed which demonstrates the ability to invoke task specific behavior in a language model with natural language.

3.7. Translation

We test whether GPT-2 has begun to learn how to translate from one language to another. In order to help it infer that this is the desired task, we condition the language model on a context of example pairs of the format `english sentence = french sentence` and then after a final prompt of `english sentence = we sample from the model with greedy decoding and use the first generated sentence as the translation`. On the WMT-14 English-French test set, GPT-2 gets 5 BLEU, which is slightly worse than a word-by-word substitution with a bilingual lexicon inferred in previous work on unsupervised word translation

由于19%的答案不在上下文中，我们使用了未经预处理的版本数据集。

3.4. 维诺格拉德模式挑战

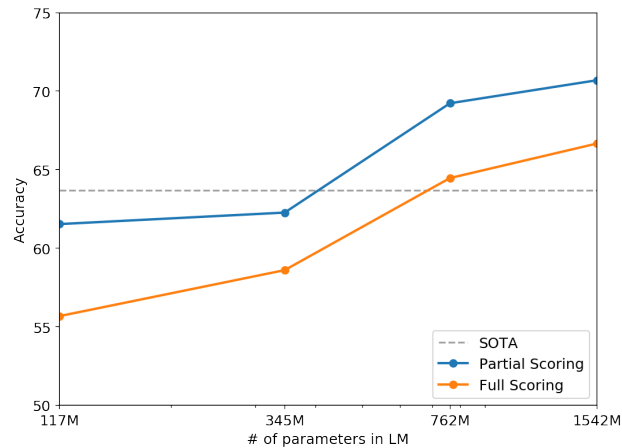


图3. 在Winograd模式挑战中，性能随模型容量的变化情况。

Winograd Schema挑战（Levesque等人，2012年）旨在通过评估系统解决文本歧义的能力来衡量其进行常识推理的能力。最近，Trinh和Le（2018年）利用语言模型在此挑战上取得了显著进展，通过以更高概率预测歧义消解来实现。我们遵循他们的问题设定，并在图3中通过完整和部分评分技术可视化我们模型的性能。GPT-2将最先进准确率提高了7%，达到70.70%。该数据集规模较小，仅包含273个示例，因此我们建议阅读Trichelair等人（2018年）的研究以帮助理解这一结果。

3.5. 阅读理解

对话式问答数据集（CoQA）Reddy等人（2018）包含来自7个不同领域的文档，每个文档都配有提问者 and 回答者之间关于该文档的自然语言对话。CoQA不仅测试阅读理解能力，还测试模型回答依赖对话历史的问题（例如“为什么？”）的能力。

当以文档、相关对话历史和最终标记A为条件时，GPT-2的贪婪解码在开发集上达到55 F1值。这相当于或超过了四分之三基线系统的性能，且未使用这些基线训练所依赖的12.7万+个手动收集的问答对。基于BERT的监督式SOTA系统（Devlin等人，

	R-1	R-2	R-L	R-AVG
Bottom-Up Sum	41.22	18.68	38.34	32.75
Lede-3	40.38	17.66	36.62	31.55
Seq2Seq + Attn	31.33	11.81	28.83	23.99
GPT-2 TL;DR:	29.34	8.27	26.58	21.40
Random-3	28.78	8.63	25.52	20.98
GPT-2 no hint	21.58	4.03	19.47	15.03

表4. 在CNN和Daily Mail数据集上通过ROUGE F1指标衡量的摘要性能。Bottom-Up Sum是来自(Gehrmann等人，2018)的SOTA模型。

2018年），已接近人类89 F1的表现。尽管GPT-2作为一个没有任何监督训练的系统，其表现令人振奋，但对其答案和错误的一些检查表明，GPT-2经常使用基于简单检索的启发式方法，例如 *answer with a name from the document in response to a who question*。

3.6. 总结

我们在CNN和Daily Mail数据集（Nallapati等人，2016）上测试了GPT-2执行摘要生成的能力。为引导摘要生成行为，我们在文章后添加文本“TL;DR:”，并使用Top- k 随机采样（Fan等人，2018）生成100个词元，其中 $k = 2$ 以减少重复并鼓励比贪婪解码更具抽象性的摘要。我们取这100个词元中生成的前3个句子作为摘要。虽然从质量上看生成结果类似摘要（如表14所示），但它们往往聚焦于文章的近期内容，或混淆具体细节（例如事故涉及多少辆汽车、标志在帽子上还是衬衫上）。在常用的ROUGE 1、2、L指标上，生成的摘要仅开始接近经典神经基线的性能，且仅略优于从文章中随机选择3个句子的方法。当移除任务提示时，GPT-2在综合指标上的性能下降了6.4分，这证明了通过自然语言在语言模型中激发特定任务行为的能力。

3.7. 翻译

我们测试GPT-2是否已开始学习如何将一种语言翻译成另一种语言。为了帮助模型推断出这是期望的任务，我们以示例对格式的上下文作为语言模型的条件，即英语句子=法语句子，然后在给出最终提示英语句子=后，我们通过贪婪解码从模型中采样，并将第一个生成的句子作为翻译结果。在WMT-14英法测试集上，GPT-2获得了5 BLEU分，这比先前无监督词翻译工作中通过双语词典逐词替换的结果略差。

Question	Generated Answer	Correct	Probability
Who wrote the book the origin of species?	Charles Darwin	✓	83.4%
Who is the founder of the ubuntu project?	Mark Shuttleworth	✓	82.0%
Who is the quarterback for the green bay packers?	Aaron Rodgers	✓	81.1%
Panda is a national animal of which country?	China	✓	76.8%
Who came up with the theory of relativity?	Albert Einstein	✓	76.4%
When was the first star wars film released?	1977	✓	71.4%
What is the most common blood type in sweden?	A	✗	70.6%
Who is regarded as the founder of psychoanalysis?	Sigmund Freud	✓	69.3%
Who took the first steps on the moon in 1969?	Neil Armstrong	✓	66.8%
Who is the largest supermarket chain in the uk?	Tesco	✓	65.3%
What is the meaning of shalom in english?	peace	✓	64.0%
Who was the author of the art of war?	Sun Tzu	✓	59.6%
Largest state in the us by land mass?	California	✗	59.2%
Green algae is an example of which type of reproduction?	parthenogenesis	✗	56.5%
Vikram samvat calender is official in which country?	India	✓	55.6%
Who is mostly responsible for writing the declaration of independence?	Thomas Jefferson	✓	53.3%
What us state forms the western boundary of montana?	Montana	✗	52.3%
Who plays ser davos in game of thrones?	Peter Dinklage	✗	52.1%
Who appoints the chair of the federal reserve system?	Janet Yellen	✗	51.5%
State the process that divides one nucleus into two genetically identical nuclei?	mitosis	✓	50.7%
Who won the most mvp awards in the nba?	Michael Jordan	✗	50.2%
What river is associated with the city of rome?	the Tiber	✓	48.6%
Who is the first president to be impeached?	Andrew Johnson	✓	48.3%
Who is the head of the department of homeland security 2017?	John Kelly	✓	47.0%
What is the name given to the common currency to the european union?	Euro	✓	46.8%
What was the emperor name in star wars?	Palpatine	✓	46.5%
Do you have to have a gun permit to shoot at a range?	No	✓	46.4%
Who proposed evolution in 1859 as the basis of biological development?	Charles Darwin	✓	45.7%
Nuclear power plant that blew up in russia?	Chernobyl	✓	45.7%
Who played john connor in the original terminator?	Arnold Schwarzenegger	✗	45.2%

Table 5. The 30 most confident answers generated by GPT-2 on the development set of Natural Questions sorted by their probability according to GPT-2. None of these questions appear in WebText according to the procedure described in Section 4.

(Conneau et al., 2017b). On the WMT-14 French-English test set, GPT-2 is able to leverage its very strong English language model to perform significantly better, achieving 11.5 BLEU. This outperforms several unsupervised machine translation baselines from (Artetxe et al., 2017) and (Lample et al., 2017) but is still much worse than the 33.5 BLEU of the current best unsupervised machine translation approach (Artetxe et al., 2019). Performance on this task was surprising to us, since we deliberately removed non-English webpages from WebText as a filtering step. In order to confirm this, we ran a byte-level language detector² on WebText which detected only 10MB of data in the French language which is approximately 500x smaller than the monolingual French corpus common in prior unsupervised machine translation research.

3.8. Question Answering

A potential way to test what information is contained within a language model is to evaluate how often it generates the correct answer to factoid-style questions. Previous showcasing of this behavior in neural systems where all information is stored in parameters such as *A Neural Conversational Model* (Vinyals & Le, 2015) reported qualitative results due to the lack of high-quality evaluation datasets. The recently introduced Natural Questions dataset (Kwiatkowski et al.,

2019) is a promising resource to test this more quantitatively. Similar to translation, the context of the language model is seeded with example question answer pairs which helps the model infer the short answer style of the dataset. GPT-2 answers 4.1% of questions correctly when evaluated by the exact match metric commonly used on reading comprehension datasets like SQUAD.³ As a comparison point, the smallest model does not exceed the 1.0% accuracy of an incredibly simple baseline which returns the most common answer for each question type (who, what, where, etc...). GPT-2 answers 5.3 times more questions correctly, suggesting that model capacity has been a major factor in the poor performance of neural systems on this kind of task as of yet. The probability GPT-2 assigns to its generated answers is well calibrated and GPT-2 has an accuracy of 63.1% on the 1% of questions it is most confident in. The 30 most confident answers generated by GPT-2 on development set questions are shown in Table 5. The performance of GPT-2 is still much, much, worse than the 30 to 50% range of open domain question answering systems which hybridize information retrieval with extractive document question answering (Alberti et al., 2019).

³Alec, who previously thought of himself as good at random trivia, answered 17 of 100 randomly sampled examples correctly when tested in the same setting as GPT-2. He actually only got 14 right but he should have gotten those other 3

²<https://github.com/CLD2Owners/cld2>

Question	Generated Answer	Correct	Probability
Who wrote the book the origin of species?	Charles Darwin	✓	83.4%
Who is the founder of the ubuntu project?	Mark Shuttleworth	✓	82.0%
Who is the quarterback for the green bay packers?	Aaron Rodgers	✓	81.1%
Panda is a national animal of which country?	China	✓	76.8%
Who came up with the theory of relativity?	Albert Einstein	✓	76.4%
When was the first star wars film released?	1977	✓	71.4%
What is the most common blood type in sweden?	A	✗	70.6%
Who is regarded as the founder of psychoanalysis?	Sigmund Freud	✓	69.3%
Who took the first steps on the moon in 1969?	Neil Armstrong	✓	66.8%
Who is the largest supermarket chain in the uk?	Tesco	✓	65.3%
What is the meaning of shalom in english?	peace	✓	64.0%
Who was the author of the art of war?	Sun Tzu	✓	59.6%
Largest state in the us by land mass?	California	✗	59.2%
Green algae is an example of which type of reproduction?	parthenogenesis	✗	56.5%
Vikram samvat calender is official in which country?	India	✓	55.6%
Who is mostly responsible for writing the declaration of independence?	Thomas Jefferson	✓	53.3%
What us state forms the western boundary of montana?	Montana	✗	52.3%
Who plays ser davos in game of thrones?	Peter Dinklage	✗	52.1%
Who appoints the chair of the federal reserve system?	Janet Yellen	✗	51.5%
State the process that divides one nucleus into two genetically identical nuclei?	mitosis	✓	50.7%
Who won the most mvp awards in the nba?	Michael Jordan	✗	50.2%
What river is associated with the city of rome?	the Tiber	✓	48.6%
Who is the first president to be impeached?	Andrew Johnson	✓	48.3%
Who is the head of the department of homeland security 2017?	John Kelly	✓	47.0%
What is the name given to the common currency to the european union?	Euro	✓	46.8%
What was the emperor name in star wars?	Palpatine	✓	46.5%
Do you have to have a gun permit to shoot at a range?	No	✓	46.4%
Who proposed evolution in 1859 as the basis of biological development?	Charles Darwin	✓	45.7%
Nuclear power plant that blew up in russia?	Chernobyl	✓	45.7%
Who played john connor in the original terminator?	Arnold Schwarzenegger	✗	45.2%

表5：由GPT-2在Natural Questions开发集上生成的30个置信度最高的答案，按GPT-2给出的概率排序。根据第4节所述流程，这些问题均未出现在WebText中。

(Conneau等人，2017b)。在WMT-14法语-英语测试集上，GPT-2能够利用其非常强大的英语语言模型显著提升表现，达到11.5 BLEU分数。这超越了(Artetxe等人，2017)和(Lample等人，2017)提出的若干无监督机器翻译基线方法，但仍远低于当前最佳无监督机器翻译方法(Artetxe等人，2019)的33.5 BLEU分数。该任务的表现令我们感到意外，因为我们曾在WebText数据预处理阶段刻意过滤了非英语网页。为验证此现象，我们对WebText运行了字节级语言检测器²，仅检测出10MB法语数据，这比先前无监督机器翻译研究中常用的单语法语语料库规模小约500倍。

3.8. 问答

测试语言模型包含何种信息的一种潜在方法是评估其生成事实型问题正确答案的频率。先前在神经网络系统中展示这种行为时，由于缺乏高质量评估数据集，所有信息都存储在参数中的研究（如 *A Neural Conversational Model* (Vinyals & Le, 2015)）仅报告了定性结果。最近引入的自然问题数据集（Kwiatkowski等人，

²<https://github.com/CLD2Owners/cld2>

2019年）是进行更定量化测试的理想资源。与翻译类似，语言模型的上下文通过示例问答对进行初始化，这有助于模型推断数据集的简短答案风格。在采用SQ UAD等阅读理解数据集常用精确匹配指标评估时，GPT-2能正确回答4.1%的问题³。作为对比，最小模型的准确率未超过1.0%——这个极简基线仅返回每种问题类型（人物、事件、地点等）的最常见答案。GPT-2的正确答题量高出5.3倍，表明模型容量迄今仍是神经网络在此类任务中表现不佳的主要因素。GPT-2为其生成答案分配的概率校准良好，在其置信度最高的1%问题上准确率达到63.1%。表5展示了GPT-2在开发集问题上生成的前30个最高置信度答案。GPT-2的性能仍远逊于开放域问答系统30%-50%的表现区间，后者融合了信息检索与抽取式文档问答技术（Alberti等人，2019年）。

³Alec, who previously thought of himself as good at random trivia, answered 17 of 100 randomly sampled examples correctly when tested in the same setting as GPT-2. He actually only got 14 right but he should have gotten those other 3

Language Models are Unsupervised Multitask Learners						
	PTB	WikiText-2	enwik8	text8	Wikitext-103	1BW
Dataset train	2.67%	0.66%	7.50%	2.34%	9.09%	13.19%
WebText train	0.88%	1.63%	6.31%	3.94%	2.42%	3.75%

Table 6. Percentage of test set 8 grams overlapping with training sets.

4. Generalization vs Memorization

Recent work in computer vision has shown that common image datasets contain a non-trivial amount of near-duplicate images. For instance CIFAR-10 has 3.3% overlap between train and test images (Barz & Denzler, 2019). This results in an over-reporting of the generalization performance of machine learning systems. As the size of datasets increases this issue becomes increasingly likely which suggests a similar phenomena could be happening with WebText. Therefore it is important to analyze how much test data also shows up in the training data.

To study this we created Bloom filters containing 8-grams of WebText training set tokens. To improve recall, strings were normalized to contain only lower-cased alphanumeric words with a single space as a delimiter. The Bloom filters were constructed such that the false positive rate is upper bounded by $\frac{1}{10^8}$. We further verified the low false positive rate by generating 1M strings, of which zero were found by the filter.

These Bloom filters let us calculate, given a dataset, the percentage of 8-grams from that dataset that are also found in the WebText training set. Table 6 shows this overlap analysis for the test sets of common LM benchmarks. Common LM datasets’ test sets have between 1-6% overlap with WebText train, with an average of overlap of 3.2%. Somewhat surprisingly, many datasets have larger overlaps with their own training splits, with an average of 5.9% overlap.

Our approach optimizes for recall, and while manual inspection of the overlaps shows many common phrases, there are many longer matches that are due to duplicated data. This is not unique to WebText. For instance, we discovered that the test set of WikiText-103 has an article which is also in the training dataset. Since there are only 60 articles in the test set there is at least an overlap of 1.6%.⁴ Potentially more worryingly, 1BW has an overlap of nearly 13.2% with its own training set according to our procedure.

For the Winograd Schema Challenge, we found only 10 schemata which had any 8-gram overlaps with the WebText training set. Of these, 2 were spurious matches. Of the remaining 8, only 1 schema appeared in any contexts that

⁴A significant portion of additional overlap is due to editors reusing some paragraphs across multiple articles with a shared theme such as various battles in the Korean War.

gave away the answer.

For CoQA, about 15% of documents in the news domain are already in WebText and the model performs about 3 F1 better on these. CoQA’s development set metric reports the average performance over 5 different domains and we measure a gain of about 0.5-1.0 F1 due to overlap across the various domains. However, no actual training questions or answers are in WebText since CoQA was released after the cutoff date for links in WebText.

On LAMBADA, the average overlap is 1.2%. GPT-2 performs about 2 perplexity better on examples with greater than 15% overlap. Recalculating metrics when excluding all examples with any overlap shifts results from 8.6 to 8.7 perplexity and reduces accuracy from 63.2% to 62.9%. This very small change in overall results is likely due to only 1 in 200 examples having significant overlap.

Overall, our analysis suggests that data overlap between WebText training data and specific evaluation datasets provides a small but consistent benefit to reported results. However, for most datasets we do not notice significantly larger overlaps than those already existing between standard training and test sets, as Table 6 highlights.

Understanding and quantifying how highly similar text impacts performance is an important research question. Better de-duplication techniques such as scalable fuzzy matching could also help better answer these questions. For now, we recommend the use of n-gram overlap based de-duplication as an important verification step and sanity check during the creation of training and test splits for new NLP datasets.

Another potential way of determining whether the performance of WebText LMs is attributable to memorization is inspecting their performance on their own held-out set. As shown in Figure 4, performance on both the training and test sets of WebText are similar and improve together as model size is increased. This suggests even GPT-2 is still underfitting on WebText in many ways.

GPT-2 is also able to write news articles about the discovery of talking unicorns. An example is provided in Table 13.

5. Related Work

A significant portion of this work measured the performance of larger language models trained on larger datasets. This

Language Models are Unsupervised Multitask Learners						
	PTB	WikiText-2	enwik8	text8	Wikitext-103	1BW
Dataset train	2.67%	0.66%	7.50%	2.34%	9.09%	13.19%
WebText train	0.88%	1.63%	6.31%	3.94%	2.42%	3.75%

表6. 测试集8-gram与训练集重叠百分比。

4. 泛化与记忆

计算机视觉领域的最新研究表明，常见的图像数据集中包含相当数量的近似重复图像。例如，CIFAR-10数据集的训练集与测试集之间存在3.3%的重叠图像（Barz & Denzler, 2019）。这导致机器学习系统的泛化性能被高估。随着数据集规模的扩大，此类问题出现的可能性也随之增加，这意味着WebText数据集可能面临类似现象。因此，分析测试数据在训练数据中的重复出现比例至关重要。

为了研究这一点，我们创建了包含WebText训练集标记的8-gram的布隆过滤器。为了提高召回率，字符串被归一化为仅包含小写字母数字单词，并以单个空格作为分隔符。布隆过滤器的构建使得误报率上限为 $\frac{1}{10^8}$ 。我们通过生成100万个字符串进一步验证了低误报率，其中过滤器未发现任何误报。

这些布隆过滤器使我们能够计算，给定一个数据集，其中有多少百分比的8-gram也出现在WebText训练集中。表6展示了常见语言模型基准测试集的这种重叠分析。常见语言模型数据集的测试集与WebText训练集有1-6%的重叠，平均重叠率为3.2%。有些令人惊讶的是，许多数据集与其自身的训练分割有更大的重叠，平均重叠率为5.9%。

我们的方法优化了召回率，尽管手动检查重叠部分显示了许多常见短语，但许多较长的匹配是由于数据重复造成的。这并非WebText独有。例如，我们发现WikiText-103的测试集中有一篇文章也出现在训练数据集中。由于测试集仅有60篇文章，至少存在1.6%的重叠率。⁴更令人担忧的是，根据我们的流程，1BW与其自身训练集的重叠率接近13.2%。

对于Winograd模式挑战，我们只发现了10个模式与WebText训练集存在任何8-gram重叠。其中，有2个是虚假匹配。在剩下的8个中，只有1个模式出现在任何上下文中

⁴A significant portion of additional overlap is due to editors reusing some paragraphs across multiple articles with a shared theme such as various battles in the Korean War.

泄露了答案。

对于CoQA，新闻领域约15%的文档已存在于WebText中，模型在这些文档上的表现提升了约3个F1值。CoQA开发集的评估指标报告了五个不同领域的平均性能，我们测得由于各领域间的重叠，性能提升了约0.5-1.0个F1值。然而，由于CoQA的发布时间晚于WebText链接的截止日期，WebText中并未包含实际的训练问题或答案。

在LAMBADA数据集上，平均重叠率为1.2%。GPT-2在重叠率超过15%的示例上表现约降低2个困惑度。若排除所有存在重叠的示例重新计算指标，困惑度从8.6变为8.7，准确率从63.2%降至62.9%。整体结果的微小变化很可能源于每200个示例中仅有1个存在显著重叠。

总体而言，我们的分析表明，WebText训练数据与特定评估数据集之间的数据重叠为报告结果带来了虽小但持续的好处。然而，对于大多数数据集，我们并未发现比标准训练集和测试集之间已有重叠显著更大的重叠，如表6所示。

理解和量化高度相似文本如何影响性能是一个重要的研究问题。更好的去重技术，如可扩展的模糊匹配，也有助于更好地回答这些问题。目前，我们建议使用基于n-gram重叠的去重方法，作为创建新NLP数据集训练和测试分割时的重要验证步骤和合理性检查。

另一种确定WebText语言模型性能是否归因于记忆的潜在方法是检查它们在自己保留集上的表现。如图4所示，WebText训练集和测试集上的表现相似，并且随着模型规模的增加而同步提升。这表明即使GPT-2在许多方面对WebText仍然存在欠拟合。

GPT-2还能撰写关于发现会说话的独角兽的新闻报道。表13中提供了一个例子。

5. 相关工作

这项工作的一个重要部分是测量在更大数据集上训练的大型语言模型的性能。这

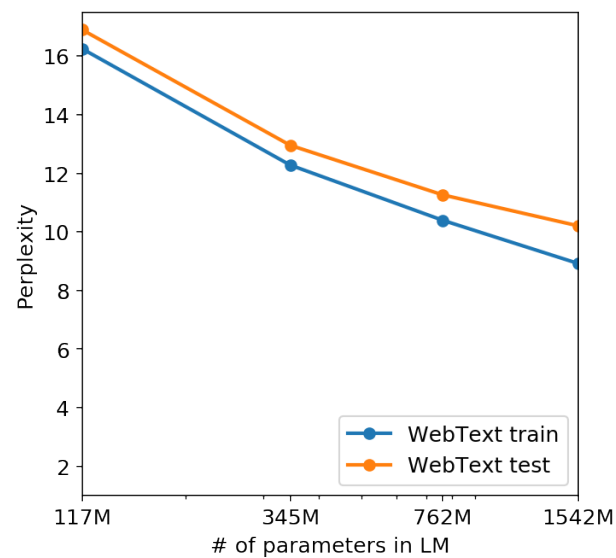


Figure 4. The performance of LMs trained on WebText as a function of model size.

is similar to the work of Jozefowicz et al. (2016) which scaled RNN based language models on the 1 Billion Word Benchmark. Bajgar et al. (2016) also previously improved results on the Children’s Book Test by creating a much larger training dataset out of Project Gutenberg to supplement the standard training dataset. Hestness et al. (2017) conducted a thorough analysis of how the performance of various deep learning models changes as a function of both model capacity and dataset size. Our experiments, while much noisier across tasks, suggest similar trends hold for sub-tasks of an objective and continue into the 1B+ parameter regime.

Interesting learned functionality in generative models has been documented before such as the cells in an RNN language model performing line-width tracking and quote/comment detection Karpathy et al. (2015). More inspirational to our work was the observation of Liu et al. (2018) that a model trained to generate Wikipedia articles also learned to translate names between languages.

Previous work has explored alternative approaches to filtering and constructing a large text corpus of web pages, such as the iWeb Corpus (Davies, 2018).

There has been extensive work on pre-training methods for language tasks. In addition to those mentioned in the introduction, GloVe (Pennington et al., 2014) scaled word vector representation learning to all of Common Crawl. An influential early work on deep representation learning for text was *Skip-thought Vectors* (Kiros et al., 2015). McCann et al. (2017) explored the use of representations derived from machine translation models and Howard & Ruder (2018)

improved the RNN based fine-tuning approaches of (Dai & Le, 2015). (Conneau et al., 2017a) studied the transfer performance of representations learned by natural language inference models and (Subramanian et al., 2018) explored large-scale multitask training.

(Ramachandran et al., 2016) demonstrated that seq2seq models benefit from being initialized with pre-trained language models as encoders and decoders. More recent work has shown that LM pre-training is helpful when fine-tuned for difficult generation tasks like chit-chat dialog and dialog based question answering systems as well (Wolf et al., 2019) (Dinan et al., 2018).

6. Discussion

Much research has been dedicated to learning (Hill et al., 2016), understanding (Levy & Goldberg, 2014), and critically evaluating (Wieting & Kiela, 2019) the representations of both supervised and unsupervised pre-training methods. Our results suggest that unsupervised task learning is an additional promising area of research to explore. These findings potentially help explain the widespread success of pre-training techniques for down-stream NLP tasks as we show that, in the limit, one of these pre-training techniques begins to learn to perform tasks directly without the need for supervised adaption or modification.

On reading comprehension the performance of GPT-2 is competitive with supervised baselines in a zero-shot setting. However, on other tasks such as summarization, while it is qualitatively performing the task, its performance is still only rudimentary according to quantitative metrics. While suggestive as a research result, in terms of practical applications, the zero-shot performance of GPT-2 is still far from use-able.

We have studied the zero-shot performance of WebText LMs on many canonical NLP tasks, but there are many additional tasks that could be evaluated. There are undoubtedly many practical tasks where the performance of GPT-2 is still no better than random. Even on common tasks that we evaluated on, such as question answering and translation, language models only begin to outperform trivial baselines when they have sufficient capacity.

While zero-shot performance establishes a baseline of the potential performance of GPT-2 on many tasks, it is not clear where the ceiling is with finetuning. On some tasks, GPT-2’s fully abstractive output is a significant departure from the extractive pointer network (Vinyals et al., 2015) based outputs which are currently state of the art on many question answering and reading comprehension datasets. Given the prior success of fine-tuning GPT, we plan to investigate fine-tuning on benchmarks such as decaNLP and GLUE, especially since it is unclear whether the additional

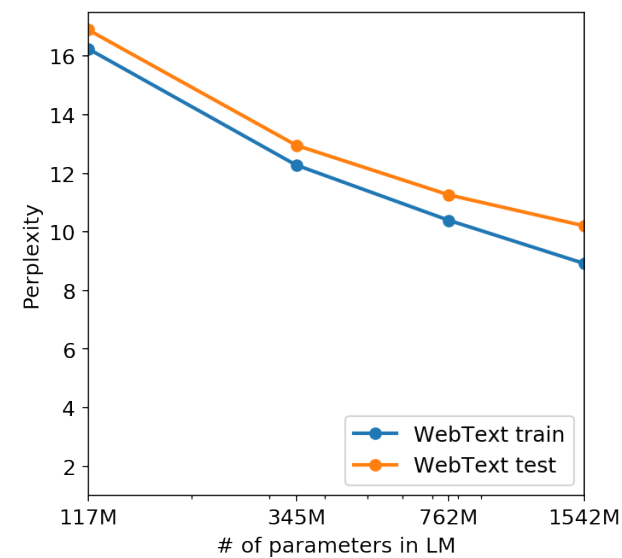


图4. 在WebText上训练的语言模型性能随模型规模的变化函数。

类似于Jozefowicz等人（2016）在10亿词基准上扩展基于RNN的语言模型的工作。Bajgar等人（2016）此前也通过利用古腾堡计划创建更大的训练数据集来补充标准训练数据，从而改进了儿童图书测试的结果。Hestness等人（2017）对多种深度学习模型的性能如何随模型容量和数据集大小的变化进行了深入分析。我们的实验虽然在各项任务中噪声更大，但表明类似趋势也适用于目标的子任务，并延续到1B+参数范围。

生成模型中一些有趣的学习功能先前已有记录，例如RNN语言模型中的单元执行行宽跟踪和引文/评论检测（Karpathy等人，2015年）。更启发我们工作的是Liu等人（2018年）的观察，即训练用于生成维基百科文章的模型也学会了在不同语言间翻译名称。

先前的研究已经探索了筛选和构建大型网页文本语料库的替代方法，例如iWeb语料库（Davies, 2018）。

在语言任务的预训练方法方面已有大量工作。除了引言中提到的那些，GloVe (Pennington et al., 2014) 将词向量表示学习扩展到了整个Common Crawl。文本深度表示学习领域一项具有影响力的早期工作是 *Skip-thought Vectors* (Kiros等人，2015年)。McCann等人（2017年）探索了从机器翻译模型衍生的表示方法的应用，而Howard & Ruder（2018年）

改进了基于RNN的(Dai & Le, 2015)微调方法。(Conneau et al., 2017a)研究了自然语言推理模型学习到的表征的迁移性能，(Subramanian et al., 2018)则探索了大规模多任务训练。

(Ramachandran等人，2016年)的研究表明，序列到序列模型通过使用预训练语言模型作为编码器和解码器进行初始化能够获益。更近期的研究进一步证实，语言模型预训练在经过微调后，对于诸如闲聊对话和基于对话的问答系统等复杂生成任务同样具有帮助(Wolf等人，2019年)(Dinan等人，2018年)。

6. 讨论

大量研究致力于学习（Hill等人，2016）、理解（Levy & Goldberg, 2014）以及批判性评估（Wieting & Kiela, 2019）有监督和无监督预训练方法的表征。我们的结果表明，无监督任务学习是另一个值得探索的潜力研究领域。这些发现可能有助于解释预训练技术在下游NLP任务中广泛成功的原因——正如我们所示，在极限情况下，其中一种预训练技术开始直接学习执行任务，而无需有监督的适应或修改。

在阅读理解方面，GPT-2在零样本设置下的表现与有监督基线模型相当。然而，在其他任务如摘要生成上，虽然它能定性执行任务，但根据定量指标，其性能仍处于初级阶段。尽管作为研究成果具有启发性，但在实际应用层面，GPT-2的零样本性能仍远未达到可用水平。

我们研究了WebText语言模型在许多经典NLP任务上的零样本表现，但还有许多额外任务有待评估。毫无疑问，在许多实际任务中，GPT-2的表现仍然不优于随机猜测。即使在我们评估的常见任务上，如问答和翻译，语言模型也只有在具备足够容量时，才开始超越简单基线。

虽然零样本性能为GPT-2在许多任务上的潜在表现建立了基准，但通过微调能达到的上限尚不明确。在某些任务中，GPT-2的完全抽象化输出与当前许多问答和阅读理解数据集上最先进的基于提取式指针网络（Vinyals等人，2015）的输出存在显著差异。鉴于先前微调GPT的成功经验，我们计划在decaNLP和GLUE等基准测试中研究微调效果，特别是考虑到目前尚不清楚额外

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training data and capacity of GPT-2 is sufficient to overcome the inefficiencies of uni-directional representations demonstrated by BERT (Devlin et al., 2018).	
7. Conclusion	
When a large language model is trained on a sufficiently large and diverse dataset it is able to perform well across many domains and datasets. GPT-2 zero-shots to state of the art performance on 7 out of 8 tested language modeling datasets. The diversity of tasks the model is able to perform in a zero-shot setting suggests that high-capacity models trained to maximize the likelihood of a sufficiently varied text corpus begin to learn how to perform a surprising amount of tasks without the need for explicit supervision. ⁵	
Acknowledgements	
Thanks to everyone who wrote the text, shared the links, and upvoted the content in WebText. Many millions of people were involved in creating the data that GPT-2 was trained on. Also thanks to all the Googlers who helped us with training infrastructure, including Zak Stone, JS Riehl, Jonathan Hseu, Russell Power, Youlong Cheng, Noam Shazeer, Solomon Boulos, Michael Banfield, Aman Gupta, Daniel Sohn, and many more. Finally thanks to the people who gave feedback on drafts of the paper: Jacob Steinhardt, Sam Bowman, Geoffrey Irving, and Madison May.	
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GPT-2的训练数据和容量足以克服BERT（Devlin等人，2018年）所展示的单向表示的低效性。	
7. 结论	
当大型语言模型在足够大且多样化的数据集上进行训练时，它能够在许多领域和数据集上表现良好。GPT-2在8个测试语言建模数据集中的7个上实现了零样本的先进性能。该模型在零样本设置下能够执行任务的多样性表明，经过训练以最大化足够多样化文本语料库似然性的高容量模型，开始学习如何执行大量任务，而无需明确的监督。 ⁵	
致谢	
感谢所有撰写文本、分享链接以及在WebText上为内容点赞的人。数以百万计的人参与了GPT-2训练数据的创建。同时感谢所有在训练基础设施方面帮助我们的谷歌员工，包括Zak Stone、JS Riehl、Jonathan Hseu、Russell Power、Youlong Cheng、Noam Shazeer、Solomon Boulos、Michael Banfield、Aman Gupta、Daniel Sohn等。最后感谢对论文草稿提供反馈的人：Jacob Steinhardt、Sam Bowman、Geoffrey Irving和Madison May。	
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8. Appendix A: Samples

8.1. Model capacity

To complement the reported perplexity gains of bigger LMs on WebText show in Figure 4, Tables 7 through 11 show side-by-side completions of the smallest WebText LM and GPT-2 on random unseen WebText test set articles.

8.2. Text Memorization

We observe some memorizing behavior in GPT-2 on longer strings that are repeated many times in the dataset such as famous quotes or speeches. For example, when conditioned on the first sentence and a half of the Gettysburg Address (which occurs approximately 40 times throughout WebText), an argmax decode from GPT-2 recovers the speech. Even when sampling without truncation, we find that the model copies the speech for awhile before drifting, albeit in a similar style. It typically drifts within 100-200 tokens, and displays widening diversity once it drifts.

To quantify how often exact memorization shows up in samples, we generated samples from GPT-2 conditioned on WebText test set articles and compared the overlap rates of GPT-2’s generations to the overlap rates of the ground-truth completions. The results of this analysis are shown below and suggest that GPT-2 repeats text from the training set less often then the baseline rate of held-out articles.

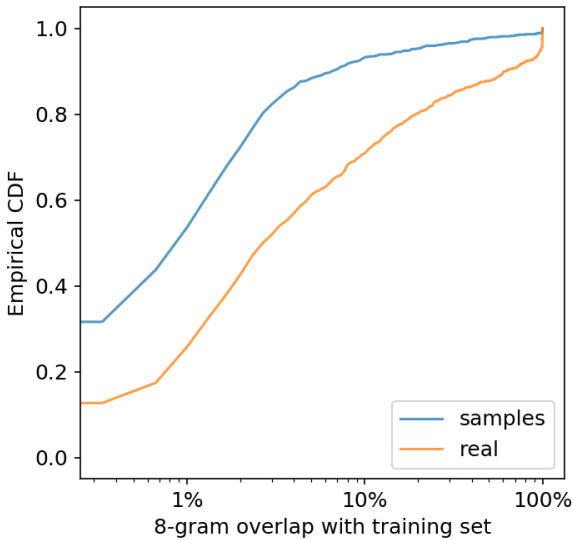


Figure 5. CDF of percentage 8-gram overlap with WebText training set, for both WebText test set and samples (conditioned on WebText test set, with top- k truncated random sampling with $k = 40$). Most samples have less than 1% overlap, including over 30% of samples with no overlap, whereas the median for test set is 2.6% overlap.

8.3. Diversity

Table 12 shows multiple completions of the same random WebText test set context, showing the diversity of completions with standard sampling settings.

8.4. Robustness

Table 13 shows the previously mentioned talking unicorns news article. We find the model to be capable of handling out of distribution contexts, but the quality of these samples is generally lower.

8. 附录A：样本

8.1. 模型容量

为了补充图4中展示的更大语言模型在WebText上报告的困惑度提升，表7至表11并列展示了最小WebText语言模型与GPT-2在随机未见过的WebText测试集文章上的补全结果。

8.2. 文本记忆

我们观察到GPT-2对数据集中重复多次的长字符串（如人名名言或演讲）存在一定的记忆行为。例如，当以葛底斯堡演说（在WebText中约出现40次）的前一句半作为条件时，GPT-2通过argmax解码能完整复现该演讲。即使在不进行截断的采样过程中，我们也发现模型会先复制一段演讲内容才开始偏离，尽管风格仍保持相似。这种偏离通常发生在100-200个标记内，且一旦偏离后多样性会显著增加。

为了量化精确记忆在样本中出现的频率，我们基于WebText测试集文章生成了GPT-2的样本，并将GPT-2生成内容的重叠率与真实补全内容的重叠率进行了比较。此分析结果如下所示，表明GPT-2重复训练集文本的频率低于预留文章的基线率。

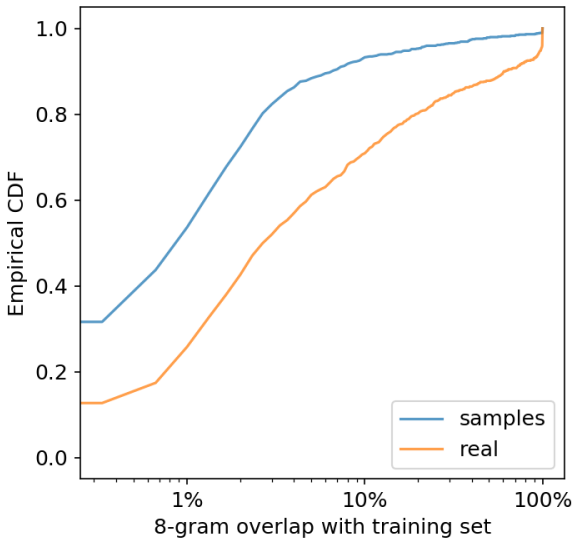


图5. 针对WebText测试集和样本（以WebText测试集为条件，采用top- k 截断随机采样， $k = 40$ ）与WebText训练集之间8-gram重叠百分比的经验累积分布函数。大多数样本的重叠率低于1%，其中超过30%的样本无重叠，而测试集的重叠率中位数为2.6%。

8.3. 多样性

表12展示了同一随机WebText测试集上下文的多种补全结果，展现了标准采样设置下补全的多样性。

8.4. 鲁棒性

表13展示了之前提到的会说话的独角兽新闻文章。我们发现模型能够处理分布外情境，但这些样本的质量通常较低。

Context (WebText test)	
Outfit 1: Typical	
This pairing was the first outfit I thought of when I bought the shoes. It’s like a summer version of this Jake Grantham outfit; in fact, my shoes are close to the colors of his Nike Racers! Instead of a heavy Harris Tweed jacket and denim shirt, I’m wearing a cotton DB jacket and and a linen shirt. Both fabrics (in these colors) are an absolute must for summer, as they go with both dark and and light pants! As you can see, they pair wonderfully with the dark jeans and shoes. It’s a pseudo menswear/prep outfit.	
Overall, this is a very casual outfit which is why I paired my sneakers with it. I’m not about wearing a full wool suit with sneakers (as GQ shows a lot) but I’m definitely open to keeping things casual, like this cotton DB. Casual fabrics are key to pulling off your sneakers in a dressed down menswear outfit. I’d even suggest to wear these sneakers with a khaki chino suit or a white linen suit. Just be sure to ditch the tie or wear a tee or polo; wearing a tie with sneakers is a bit too much	
Small model completion	GPT-2 completion
.	for me!
Outfit 2: High End	Outfit 2: Casual Jacket
This outfit was just in the eye of the storm. It’s a hybrid of what ASICS does with streetwear (like J. Crew) and what J. Crew does with casual pieces that are a bit more traditional. Again, as I mentioned before, it’s a low budget item and they can be very expensive. These will definitely be an issue if you’re trying to work as a casual sneakermaker, and that’s OK but don’t hesitate if you want to try the lower price versions of their outfits!	I know that some people might be opposed to wearing sneakers in a turtleneck, but I wanted to be true to myself, so I went with a slim fitting turtleneck in a color more similar to my favorite color of the day. Overall, it’s a classic turtleneck, with nothing too flashy. I did swap my shoes for these ASICS Tiger Killshots and I wish I hadn’t did that. The shoe itself is very comfortable, however, I found that wearing them in these sneakers made my feet look a bit larger. I’d wear them with denim, chinos, whatever, but would never recommend wearing them alone. There are just too many variables involved.
Outfit 3: High Light	Outfit 3: Dress
This was a ”high light” look while I had a chance to get some color. I think the color is based on	This is the final outfit I wore the ASICS Tiger Killshots. This version of the ASICS is a much more modern and streamlined version. The shoe has the most modern and minimal design that I can think of! With a lot of black, these sneakers look very cool. They’ll definitely be very comfortable for summer.
	To me, wearing them with a suit is the easiest way to have a good pair of sneaker shoes. That’s pretty much it! If

Table 7. Random unseen contexts (top), and non-cherry-picked completions from both the smallest (left) and largest (right) models. Contexts are 768 tokens, with approximately 256 tokens worth of paragraphs shown. Completions are 256 tokens and fully shown. Top- k random sampling with $k = 40$ was used for generation.

Context (WebText test)	
Outfit 1: Typical	
This pairing was the first outfit I thought of when I bought the shoes. It’s like a summer version of this Jake Grantham outfit; in fact, my shoes are close to the colors of his Nike Racers! Instead of a heavy Harris Tweed jacket and denim shirt, I’m wearing a cotton DB jacket and and a linen shirt. Both fabrics (in these colors) are an absolute must for summer, as they go with both dark and and light pants! As you can see, they pair wonderfully with the dark jeans and shoes. It’s a pseudo menswear/prep outfit.	
Overall, this is a very casual outfit which is why I paired my sneakers with it. I’m not about wearing a full wool suit with sneakers (as GQ shows a lot) but I’m definitely open to keeping things casual, like this cotton DB. Casual fabrics are key to pulling off your sneakers in a dressed down menswear outfit. I’d even suggest to wear these sneakers with a khaki chino suit or a white linen suit. Just be sure to ditch the tie or wear a tee or polo; wearing a tie with sneakers is a bit too much	
Small model completion	GPT-2 completion
.	for me!
Outfit 2: High End	Outfit 2: Casual Jacket
This outfit was just in the eye of the storm. It’s a hybrid of what ASICS does with streetwear (like J. Crew) and what J. Crew does with casual pieces that are a bit more traditional. Again, as I mentioned before, it’s a low budget item and they can be very expensive. These will definitely be an issue if you’re trying to work as a casual sneakermaker, and that’s OK but don’t hesitate if you want to try the lower price versions of their outfits!	I know that some people might be opposed to wearing sneakers in a turtleneck, but I wanted to be true to myself, so I went with a slim fitting turtleneck in a color more similar to my favorite color of the day. Overall, it’s a classic turtleneck, with nothing too flashy. I did swap my shoes for these ASICS Tiger Killshots and I wish I hadn’t did that. The shoe itself is very comfortable, however, I found that wearing them in these sneakers made my feet look a bit larger. I’d wear them with denim, chinos, whatever, but would never recommend wearing them alone. There are just too many variables involved.
Outfit 3: High Light	Outfit 3: Dress
This was a ”high light” look while I had a chance to get some color. I think the color is based on	This is the final outfit I wore the ASICS Tiger Killshots. This version of the ASICS is a much more modern and streamlined version. The shoe has the most modern and minimal design that I can think of! With a lot of black, these sneakers look very cool. They’ll definitely be very comfortable for summer.
	To me, wearing them with a suit is the easiest way to have a good pair of sneaker shoes. That’s pretty much it! If

Table 7. Random unseen contexts (top), and non-cherry-picked completions from both the smallest (left) and largest (right) models. Contexts are 768 tokens, with approximately 256 tokens worth of paragraphs shown. Completions are 256 tokens and fully shown. Top- k random sampling with $k = 40$ was used for generation.

Context (WebText test) Some of the most glorious historical attractions in Spain date from the period of Muslim rule, including The Mezquita, built as the Great Mosque of Córdoba and the Medina Azahara, also in Córdoba and now in ruins but still visitable as such and built as the Madinat al-Zahra, the Palace of al-Andalus; and the Alhambra in Granada, a splendid, intact palace. There are also two synagogues still standing that were built during the era of Muslim Spain: Santa María la Blanca in Toledo and the Synagogue of Córdoba, in the Old City. Reconquista and Imperial era [edit]	
Small model completion	GPT-2 completion
The era of Muslim Spain is very different from that of other parts of the world, with a rather extensive medieval history. While the Spanish Empire existed during the second half of the 9th century BC it was quite slowly transformed to a small, secularized empire by 1150 A.D., in a much more prosperous and diverse state. In 1050 CE the emperor of Naples, at least partially in characterised by a series of wars and invasions from the north-western coast to the western coast of the region (e.g., on the Danube River which extended all the way from Gibraltar into the south of Spain, but was quickly destroyed on the same day by the Roman army), assumed the title of Emperor of Naples. He made a large contribution over the last few decades to restore and maintain the empire, the first of which to the west. His successors were Ferdinand, who took power from the Papal Throne in 1148, while the first imperial successor was King Alfred II of Sweden. The empire was again made one of the few independent states in the European Continent and although no more than 200 million people lived on its territory during this period the people had much to say to the Emperor. From 1720 on the Muslims made up half the population of Spain.	The Reconquista (literally, "The return to the country") of 1092 and 1095 saw the establishment in the Iberian peninsula of a united kingdom of Castile and Aragon in which a Muslim dynasty would exist. This was the last time in Europe where Muslim rule would be considered "native" and the last occasion where Christian-inflamed and organized conflicts would make themselves known in Europe. The reign of Muslim Moors in Spain is often referred to, especially among non-Muslims, as the "Conquest of Granada" and Spain is sometimes referred to as Spain in "Granada". The last years of Muslim rule were followed by the Spanish Reconquista where the Muslims were driven or compelled out of the Iberian Peninsula with the victory of Ferdinand IX. in the Battle of Poitiers in March 1189 and the conquest of Jerusalem in 1187 by Saladin. The Reconquista period in Spain came to an end when the Moors retreated north. However the Moorish rule would not be fully replaced by the Christian rule until the Reconquista era resumed when the Reconquista lasted from 1229 to 1492. The last Muslim state in Spain, Granada, was conquered by the

Table 8. Random unseen contexts (top), and non-cherry-picked completions from both the smallest (left) and largest (right) models. Contexts are 768 tokens, with approximately 256 tokens worth of paragraphs shown. Completions are 256 tokens and fully shown. Top-*k* random sampling with *k* = 40 was used for generation.

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Language Models are Unsupervised Multitask Learners	
Context (WebText test) Pinky The pink ghost’s AI is designed to ”feel” opposite of the red ghost’s behavior. Pinky actually attempts to get out in front of Pac-Man. This is accomplished by setting the target 4 tiles ahead of Pac-Man’s current location in the direction that Pac-Man is travelling. One exception to this is when Pac-Man is traveling up. Due to an overflow bug in the code, the calculation includes a left offset equal to the expected up offset. The use of Pac-Man’s expected position allows the player to exploit the navigation code for Pinky. This was most often seen in situations where Pac-Man and Pinky are in close proximity to each other. For example, if they were headed toward one another, once they are less than 4 tiles apart, the target tile is now behind Pinky. If he encounters an intersection while in this configuration, he will attempt to double-back (by turning) in order to reach the target tile. Player’s would often perform what was referred to as a ”head fake” by changing directions rapidly just as Pinky was approaching an intersection. By displacing the target tile just as Pinky	
Small model completion was coming to the left, they would get the ”correct” direction. If the location is actually farther from what Pac-Man was at, the player’s will not respond to the ghost’s movement, and the ghost will instead return to the player’s current location. This is because Pac-Man does not move by turning. The ghost will have to go to another location of some kind to return where he began, and this is achieved in many ways. It will often simply leave the location, as for example if Pac-Man did jump off a cliff, he would return to the spot where he was before vanishing into space. There is a good number of possibilities these ghost paths may be possible, but the most obvious ones are if the player wants to avoid getting caught up in the game, with a goal as easy as keeping Pac-Man in the game for 1/3 as long as he was. Chasing In Pac-Man’s case, this would be similar to how Pac-Man can ”catch up” in a fight. A ghost could chase through enemies. This is useful for many reasons, both for Pac-Man and his AI. Unlike most Ghost games, this ghost simply travels in the direction from	GPT-2 completion was about to cross the intersection, Pac-Man would be able to dodge the ghosts’ projectiles and return to the safe location of the safe house. Buster The yellow ghost’s AI is specifically designed to try to avoid Pac-Man from leaving the safe house. The player must press the SPACE bar to reveal the safe house’s doors, then go into them to find the yellow ghost hiding in the next room. The player must also note that the ghost will not attempt to exit the safe house if it sees Pac-Man already there. In fact, the ghost will not attempt an exit in the first place unless they’re already in the safe house. These two behaviors are based on the fact that the player does not always attempt to reach the safe house before Pac-Man (or Blinky). The player must also beware of this glitch if a ghost (or another player) is already in the safe house and attempting to get out as soon as Pac-Man or Blinky is gone. At this point, the ghost will attempt to exit the safe house in a similar manner as it is in ”Chase” mode. Pac-Man There are some ghosts that follow Pac-Man. These ghosts are

Table 9. Random unseen contexts (top), and non-cherry-picked completions from both the smallest (left) and largest (right) models. Contexts are 768 tokens, with approximately 256 tokens worth of paragraphs shown. Completions are 256 tokens and fully shown. Top-*k* random sampling with *k* = 40 was used for generation.

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Context (WebText test)	
Corporal Michael P. Goeldin was an unskilled laborer from Ireland when he enlisted in Company A in November 1860. Goldein survived the war. Corporal Patrick O’Neal, also from Ireland, first enlisted in 1854 and served with Company L, 3d U.S. Artillery, in Oregon. He returned to the East Coast and enlisted in the company in 1860. O’Neal served until 1874, when he was named superintendent of the National Cemetery at Willets Point, New York. Corporal Benjamin Browne was a shoemaker from Orange County, New York. In August 1862, he enlisted in the newly formed 124th New York Volunteers, and was one of sixty-one men who transferred into Company A that October. Browne reenlisted in the company in February 1864 while it was camped at Brandy Station. He returned to civilian life after completing his enlistment in 1867.	
On 10 June, Artificer William Collins was promoted to corporal, probably to fill a combat leadership void for the crossing of the James River. Collins’s service record does not reflect the qualities he demonstrated to earn this promotion, but he had obviously overcome some serious problems. Born in Sacketts Harbor, New York, Collins enlisted in the company in December 1853 at the age of twenty-two, and reenlisted in December 1858. Just a month before the war began in April 1861, Collins went ”over the hill” and was not caught until three years later. Returned to the company on 22 March 1864, he was tried	
Small model completion	GPT-2 completion
on military charges and was discharged from the U.S. Military. He returned to the company, and was assigned to Division of Internal Affairs during the Civil War.	in June of that year. Sentenced to be shot for desertion, he remained on active duty.
As an engineer, Collins also had training in engineering from his native New York. After graduate studies in the Union College of Engineering, he was employed in the United States Army for two years at the same time. There he worked under Captain Daniel Pfeiffer and was promoted to colonel in May 1870, three years after Captain Pfeiffer’s enlistment. Collins returned home to England after completing his first year in the First Division to serve the next two years in the Army Corps of Engineers, and by his second year, he began to experience his service duty.ı—endoftext—ıMOSCOW, July 5 (Reuters) - U.K.-based Russian state media and government sources and experts say Moscow has sought to downplay the ”military buildup” of the Syrian opposition while maintaining support for President Donald Trump’s plan to build a wall along the U.S.-Mexico border, calling it a ”military coup”.	Captain James J. Dall was born in Virginia in 1829 during the Revolution and joined his father in the Army in 1836. He served with the 2d Virginia Infantry until 1845, when he went to the Western Reserve Regiment; and, on his discharge, he reenlisted with Company I, 25th New York Volunteers (which in 1845 numbered about 4,500 men). One of his brothers, James, served with his company in the same capacity in the same brigade. While the regiment remained in Connecticut through 1846, Captain Dall served with Company I at various times before his discharge. A veteran of the Mexican War, he had served two years on the staff of Brigadier General John J. Sullivan at San Antonio. During the Peninsula Campaign of 1863, he commanded a company of twenty-three and was in charge of the battery of fifty-four heavy mortars and the gunboat Pinta of the Battery, a gunboat and several other boats. Captain Dall was on active duty with Company I in the fall of 1865. Two years later, he was promoted to second lieutenant.
U.S.-led Syria war veteran and Syria envoy Michael Flynn (R), right, gestures with U.S.-based Syrian political analyst Ahmet Koryat, with	Captain John A. O’

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表10：随机未见上下文（顶部），以及来自最小（左）和最大（右）模型的非精选补全。上下文为768个标记，显示约256个标记的段落。补全为256个标记并完整显示。生成时使用了Top- k 随机抽样， $k = 40$ 。

Language Models are Unsupervised Multitask Learners	
Context (WebText test) Xmonad, Xresources, and Fonts Xresources can be an absolute pain (they were for me). So, this a bit of help: First of all, the way that fonts are generally handled in Xmonad is through XFT . Xft is the X FreeType Interface library and is a library that handles the actual rendering of fonts in Xmonad/urxvt (Xresources specifies a bunch of resources to things that launch under the X window manager. Settings for individual programs are specified in Xresources like so: Urxvt.background:background settins here). Anyway, fonts are set using this syntax in Xresources : urxvt*font:xft:your font name goes here:size=the size of the font goes here,xft:fall-back font 1 goes here So, thats the most basic part of fonts in urxvt specified via Xresources. Fixing /.ssh/ Permissions in Cygwin In Cygwin on Windows I found that I could not set the permissions of my /.ssh/ folder to be 0600, as is required for ssh to allow you to use keys. The symptom I had was that I no matter what I did, it always modified the owner and the group	
Small model completion was set to /.ssh/ , which was the way things were. This could not be fixed completely. I found I could make that much more workable with just typing the username with your mouse then trying to set permissions to be "0600" for the group, which is what my system is now configured to do. However, as of Cygwin 7 on Linux I have found that doing so is now "unnecessary" for my system! I also installed a package called chroot. It does that by allowing you to install/upgrade to your system directory. So, as of Windows 8 and up, installing that can require chroot, so I tried and tried and found that it wasn't so bad when installing the package on Linux and then uninstalling it. As of my last write-up on Cygwin, the script just does its job (the sudo commands do the rest), if you try it you will have to log the change. If that wasn't confusing enough, this is what I tried and just went back to my /.ssh/ directory and the chroot script. I found I was finally running out of permissions (because i forgot to create a symlink to ssh from the file system and	GPT-2 completion , making these files read-only. I tried just starting ssh and using a program like passwd which modifies the owner/group to 0644 permissions, but it didn't work. It was annoying but not nearly as bad as it could be, just annoying to have to fiddle with permissions. So I decided to do this as the reverse: For each .ssh folder in your Cygwin directory, set the permission of each file and directory to 2644. Then start ssh with: ssh-add /.ssh/ The first time you connect, your permissions of the /.ssh folder that you just had in Cygwin will be set to 0600. Then every time you connect again, your permissions will change to 0644. Makefile for Xresources as used in .xsessionrc These are made in my .xsessionrc : .xprofile: .xsession-errors: color="" For example: .xprofile: color=blue As the colors are stored in a separate file, the color in your .xsession-errors will look like cyan instead of blue. This is mostly because, as of now, only the color is assigned to your

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Language Models are Unsupervised Multitask Learners			
Context (WebText test) Aaron loves mint chocolate cake, but he requires that it be paired with mini chocolate chips, so I threw some of those in between the layers. I also had a few Peppermint Jo Jos on hand so I crushed them up and threw some of those in along with some crushed meringue cookies because, why not? It’s a total smorgasbord of minty chocolate chippy cookie crunchy goodness. I didn’t measure how much of each topping I used, but after I tasted the finished product, I wish I had added more. You can add anything you want- crushed candy canes, peppermint bark, etc. And don’t be afraid to use a heavy hand. Texture = good. If you don’t have 7-inch cake pans, you can get 3 shorter 8-inch layers out of this			
GPT-2 Completion 1 recipe. I made the longer version, then used the pan for a layer of buttercream frosting and stuck 1/3 of it into the fridge for a frosting topper. Just make sure you have all of this frosting prepared earlier than you make this cake and you won’t have to wait on it as your egg whites sit on the bottom of your mixing bowl. You want the egg whites whipped to stiff peaks? Then put your cake dish over a pot of simmering water and heat it up until it’s just barely warm. It doesn’t take more than 3-5 minutes; just enough to get it started	GPT-2 Completion 2 recipe. If you do, and you know how to make that cake, then your life will never be the same. I have two very simple rules when it comes to cookie crumbs or chocolate chips. 1) When I buy raw sugar, I do not use it until it’s cooled down and has at least half a teaspoon of sugar on it. That’s my rule. And 2) If I have a recipe that calls for something and I want to use it in a recipe, I do not substitute it with something else on-line. Like this. I found this really cute card with cute little kittens on	GPT-2 Completion 3 recipe, which will likely yield about the same amount of cake. Chocolate Buttermilk Cake Print Chunks of chocolate cake, coated in buttermilk and topped with crunchy chopped chocolate, drizzled with chocolate sauce. Simple but a nice change from chocolate cake! Author: Momma Made Ingredients For the chocolate buttermilk cake 1 1/2 cup unsalted butter , room temperature 1 1/2 cup granulated sugar 3 large eggs plus 1 egg yolk For the chocolate glaze 1/3 cup cocoa powder 1 3/4 cups powdered sugar 6 ounces semis	
GPT-2 Completion 4 recipe. Meringue Cookies 1 3/4 cup butter, softened 1 cup sugar 1 egg yolk 3 T. heavy cream 3 1/2-4 cups flour pinch of salt Peppermint-Jojo Topping 1 1/4 cups powdered sugar 1 cup chopped pecans 1/2 cup finely chopped mint leaves 1/2 cup chopped fresh mint (about 1/2 sheet) 1 tsp. peppermint extract 1/4 tsp. sea salt pinch of sea	GPT-2 Completion 5 recipe. My 2 failures at Christmas morning were not due to poor frosting skills. That was another success story. The cake was beautifully moist, and well, just a beautiful cake. It’s got very little air in it or moisture, but all the chocolate and mint was perfectly blended together. One thing that made me cringe a bit was that all those ingredients-egg yolks, sugar, melted butter, chocolate chips, melted chocolate, and mint- had to be mixed right into the cake batter, which is super time-consuming. A friend suggested that if I put the batter in a separate bowl, I	GPT-2 Completion 6 recipe, which will give you about 6-9 servings of delicious chocolate cake. To prevent your layers from sticking together, I used my 8-inch round tart pan, lined it with parchment paper, then cut up 4 7-inch round cakes from it and sandwiched the layers between them (this also worked well for the 1-layer chocolate cake as well). When you’re ready to bake this cake, remove the parchment paper, preheat oven to 350 degrees and bake for about 30 – 40 minutes, until the top is golden and a tester comes out clean. Chocolate Buttermilk Layer Cake with Peppermint Jo	

Table 12. Non-cherry-picked completions from GPT-2 generated from the same context (from WebText test). Context is 384 tokens (shown truncated), and generations are 128 tokens. Top- k random sampling with $k = 40$ was used for generation.

Language Models are Unsupervised Multitask Learners			
Context (WebText test) Aaron loves mint chocolate cake, but he requires that it be paired with mini chocolate chips, so I threw some of those in between the layers. I also had a few Peppermint Jo Jos on hand so I crushed them up and threw some of those in along with some crushed meringue cookies because, why not? It’s a total smorgasbord of minty chocolate chippy cookie crunchy goodness. I didn’t measure how much of each topping I used, but after I tasted the finished product, I wish I had added more. You can add anything you want- crushed candy canes, peppermint bark, etc. And don’t be afraid to use a heavy hand. Texture = good. If you don’t have 7-inch cake pans, you can get 3 shorter 8-inch layers out of this			
GPT-2 Completion 1 recipe. I made the longer version, then used the pan for a layer of buttercream frosting and stuck 1/3 of it into the fridge for a frosting topper. Just make sure you have all of this frosting prepared earlier than you make this cake and you won’t have to wait on it as your egg whites sit on the bottom of your mixing bowl. You want the egg whites whipped to stiff peaks? Then put your cake dish over a pot of simmering water and heat it up until it’s just barely warm. It doesn’t take more than 3-5 minutes; just enough to get it started	GPT-2 Completion 2 recipe. If you do, and you know how to make that cake, then your life will never be the same. I have two very simple rules when it comes to cookie crumbs or chocolate chips. 1) When I buy raw sugar, I do not use it until it’s cooled down and has at least half a teaspoon of sugar on it. That’s my rule. And 2) If I have a recipe that calls for something and I want to use it in a recipe, I do not substitute it with something else on-line. Like this. I found this really cute card with cute little kittens on	GPT-2 Completion 3 recipe, which will likely yield about the same amount of cake. Chocolate Buttermilk Cake Print Chunks of chocolate cake, coated in buttermilk and topped with crunchy chopped chocolate, drizzled with chocolate sauce. Simple but a nice change from chocolate cake! Author: Momma Made Ingredients For the chocolate buttermilk cake 1 1/2 cup unsalted butter , room temperature 1 1/2 cup granulated sugar 3 large eggs plus 1 egg yolk For the chocolate glaze 1/3 cup cocoa powder 1 3/4 cups powdered sugar 6 ounces semis	
GPT-2 Completion 4 recipe. Meringue Cookies 1 3/4 cup butter, softened 1 cup sugar 1 egg yolk 3 T. heavy cream 3 1/2-4 cups flour pinch of salt Peppermint-Jojo Topping 1 1/4 cups powdered sugar 1 cup chopped pecans 1/2 cup finely chopped mint leaves 1/2 cup chopped fresh mint (about 1/2 sheet) 1 tsp. peppermint extract 1/4 tsp. sea salt pinch of sea	GPT-2 Completion 5 recipe. My 2 failures at Christmas morning were not due to poor frosting skills. That was another success story. The cake was beautifully moist, and well, just a beautiful cake. It’s got very little air in it or moisture, but all the chocolate and mint was perfectly blended together. One thing that made me cringe a bit was that all those ingredients-egg yolks, sugar, melted butter, chocolate chips, melted chocolate, and mint- had to be mixed right into the cake batter, which is super time-consuming. A friend suggested that if I put the batter in a separate bowl, I	GPT-2 Completion 6 recipe, which will give you about 6-9 servings of delicious chocolate cake. To prevent your layers from sticking together, I used my 8-inch round tart pan, lined it with parchment paper, then cut up 4 7-inch round cakes from it and sandwiched the layers between them (this also worked well for the 1-layer chocolate cake as well). When you’re ready to bake this cake, remove the parchment paper, preheat oven to 350 degrees and bake for about 30 – 40 minutes, until the top is golden and a tester comes out clean. Chocolate Buttermilk Layer Cake with Peppermint Jo	

表12. 来自同一上下文（取自WebText测试集）的GPT-2非精选补全示例。上下文为384个词元（此处显示截断），生成内容为128个词元。生成时采用Top- k 随机采样方法， $k = 40$ 参数设为40。

Context (human-written): In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.
GPT-2: The scientist named the population, after their distinctive horn, Ovid’s Unicorn. These four-horned, silver-white unicorns were previously unknown to science. Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved. Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow. Pérez and the others then ventured further into the valley. “By the time we reached the top of one peak, the water looked blue, with some crystals on top,” said Pérez. Pérez and his friends were astonished to see the unicorn herd. These creatures could be seen from the air without having to move too much to see them – they were so close they could touch their horns. While examining these bizarre creatures the scientists discovered that the creatures also spoke some fairly regular English. Pérez stated, “We can see, for example, that they have a common ‘language,’ something like a dialect or dialectic.” Dr. Pérez believes that the unicorns may have originated in Argentina, where the animals were believed to be descendants of a lost race of people who lived there before the arrival of humans in those parts of South America. While their origins are still unclear, some believe that perhaps the creatures were created when a human and a unicorn met each other in a time before human civilization. According to Pérez, “In South America, such incidents seem to be quite common.” However, Pérez also pointed out that it is likely that the only way of knowing for sure if unicorns are indeed the descendants of a lost alien race is through DNA. “But they seem to be able to communicate in English quite well, which I believe is a sign of evolution, or at least a change in social organization,” said the scientist.

Table 13. Conditional generation on an out-of-distribution context by GPT-2. Cherry pick of 10 samples generated with $k = 40$.

Context (human-written): In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.
GPT-2: The scientist named the population, after their distinctive horn, Ovid’s Unicorn. These four-horned, silver-white unicorns were previously unknown to science. Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved. Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow. Pérez and the others then ventured further into the valley. “By the time we reached the top of one peak, the water looked blue, with some crystals on top,” said Pérez. Pérez and his friends were astonished to see the unicorn herd. These creatures could be seen from the air without having to move too much to see them – they were so close they could touch their horns. While examining these bizarre creatures the scientists discovered that the creatures also spoke some fairly regular English. Pérez stated, “We can see, for example, that they have a common ‘language,’ something like a dialect or dialectic.” Dr. Pérez believes that the unicorns may have originated in Argentina, where the animals were believed to be descendants of a lost race of people who lived there before the arrival of humans in those parts of South America. While their origins are still unclear, some believe that perhaps the creatures were created when a human and a unicorn met each other in a time before human civilization. According to Pérez, “In South America, such incidents seem to be quite common.” However, Pérez also pointed out that it is likely that the only way of knowing for sure if unicorns are indeed the descendants of a lost alien race is through DNA. “But they seem to be able to communicate in English quite well, which I believe is a sign of evolution, or at least a change in social organization,” said the scientist.

表13. 条件生成 GPT-2在分布外上下文上的离子化。精选10个样本 使用 $k = 40$ 生成的样本。

Language Models are Unsupervised Multitask Learners	
Article: Prehistoric man sketched an incredible array of prehistoric beasts on the rough limestone walls of a cave in modern day France 36,000 years ago. Now, with the help of cutting-edge technology, those works of art in the Chauvet-Pont-d’Arc Cave have been reproduced to create the biggest replica cave in the world. The manmade cavern named the Caverne du Pont-d’Arc has been built a few miles from the original site in Vallon-Pont-D’arc in Southern France and contains 1,000 painstakingly-reproduced drawings as well as around 450 bones and other features... Cavemen and women sketched an incredible array of prehistoric beasts on the rough limestone walls of a cave 36,000 years ago and now a replica has been created (pictured) ...	
GPT-2: The original site in Vallon-Pont-D’arc in Southern France is a Unesco World Heritage site and is the oldest known and the best preserved cave decorated by man. The replica cave was built a few miles from the original site in Vallon-Pont-D’Arc in Southern France. The cave contains images of 14 different species of animals including woolly rhinoceros, mammoths, and big cats.	Reference: Cave mimics famous Caverne du Pont-d’Arc in France, the oldest cave decorated by man and the best preserved. The replica contains all 1,000 paintings which include 425 such as a woolly rhinoceros and mammoths. Minute details were copied using 3D modelling and anamorphic techniques, often used to shoot widescreen images. The modern cave also includes replica paw prints of bears, bones and details preserved in the original cave.
Article: ‘If you can find my submarine, it’s yours,’ Russian oil billionaire Roman Abramovich once said. And it seems the oligarch and Chelsea FC owner, whose submarine is just one of the extras that came with his £300million superyacht Eclipse (perfect for getting to shore undetected), is not the only wealthy businessman splashing out on underwater exploration. Dubbed Earth’s real ‘final frontier’, the oceans are still so little-explored that billionaires are queuing up to buy vessels that give them a glimpse of the dramatic seascapes and incredible wildlife of the world’s oceans. So if you have a spare few million in the bank and want some holiday snaps and Instagram posts that will really trump everyone else, you may want to snap up one of these... Whale of a time: The OrcaSub takes you 2000 feet beneath the surface in two pressurised Perspex viewing domes for optimum exploration ...	
GPT-2: The U-Boat Worx submarine can dive to 984 feet (300 metres) and is capable of diving to the bottom of the ocean. It can be used for diving, snorkelling and even snorkel diving. The submarine is also equipped with a snorkelling platform.	Reference: Feel like a James Bond villain with the most advanced submerged ocean vehicles. Some convert from a yacht into a submarine, and others detached off for a speedy exploration. Oliver’s Travels offers a Mile Low package on its submarine, complete with chef and butler for £175,000 a night.
Article: Amina Ali Qassim is sitting with her youngest grandchild on her lap, wiping away tears with her headscarf. Only a few months old, this is the baby girl whose ears she desperately tried to cover the night the aerial bombardment started. She lay awake, she says, in a village mosque on the Yemeni island of Birim, counting explosions as the baby cried. It could have been worse though. They could have still been in their house when the first missile landed. ”Our neighbor shouted to my husband ’you have to leave, they’re coming.’ And we just ran. As soon as we left the house, the first missile fell right by it and then a second on it. It burned everything to the ground,” Qassim tells us ...	
GPT-2: Yemen is in the middle of a civil war. Saudi Arabia is leading the coalition bombing campaign. It’s been bombing Yemen for more than two months now.	Reference: Amina Ali Qassim’s family sought shelter in a mosque before fleeing Yemen. Thousands like them are boarding boats to sail to Djibouti. Saudi Arabia has been pounding Yemen in a bid to defeat Houthi rebels.

Table 14. Summaries generated by GPT-2 and reference summaries on the test set of the CNN and Daily Mail dataset.

Language Models are Unsupervised Multitask Learners	
Article: Prehistoric man sketched an incredible array of prehistoric beasts on the rough limestone walls of a cave in modern day France 36,000 years ago. Now, with the help of cutting-edge technology, those works of art in the Chauvet-Pont-d’Arc Cave have been reproduced to create the biggest replica cave in the world. The manmade cavern named the Caverne du Pont-d’Arc has been built a few miles from the original site in Vallon-Pont-D’arc in Southern France and contains 1,000 painstakingly-reproduced drawings as well as around 450 bones and other features... Cavemen and women sketched an incredible array of prehistoric beasts on the rough limestone walls of a cave 36,000 years ago and now a replica has been created (pictured) ...	
GPT-2: The original site in Vallon-Pont-D’arc in Southern France is a Unesco World Heritage site and is the oldest known and the best preserved cave decorated by man. The replica cave was built a few miles from the original site in Vallon-Pont-D’Arc in Southern France. The cave contains images of 14 different species of animals including woolly rhinoceros, mammoths, and big cats.	Reference: Cave mimics famous Caverne du Pont-d’Arc in France, the oldest cave decorated by man and the best preserved. The replica contains all 1,000 paintings which include 425 such as a woolly rhinoceros and mammoths. Minute details were copied using 3D modelling and anamorphic techniques, often used to shoot widescreen images. The modern cave also includes replica paw prints of bears, bones and details preserved in the original cave.
Article: ‘If you can find my submarine, it’s yours,’ Russian oil billionaire Roman Abramovich once said. And it seems the oligarch and Chelsea FC owner, whose submarine is just one of the extras that came with his £300million superyacht Eclipse (perfect for getting to shore undetected), is not the only wealthy businessman splashing out on underwater exploration. Dubbed Earth’s real ‘final frontier’, the oceans are still so little-explored that billionaires are queuing up to buy vessels that give them a glimpse of the dramatic seascapes and incredible wildlife of the world’s oceans. So if you have a spare few million in the bank and want some holiday snaps and Instagram posts that will really trump everyone else, you may want to snap up one of these... Whale of a time: The OrcaSub takes you 2000 feet beneath the surface in two pressurised Perspex viewing domes for optimum exploration ...	
GPT-2: The U-Boat Worx submarine can dive to 984 feet (300 metres) and is capable of diving to the bottom of the ocean. It can be used for diving, snorkelling and even snorkel diving. The submarine is also equipped with a snorkelling platform.	Reference: Feel like a James Bond villain with the most advanced submerged ocean vehicles. Some convert from a yacht into a submarine, and others detached off for a speedy exploration. Oliver’s Travels offers a Mile Low package on its submarine, complete with chef and butler for £175,000 a night.
Article: Amina Ali Qassim is sitting with her youngest grandchild on her lap, wiping away tears with her headscarf. Only a few months old, this is the baby girl whose ears she desperately tried to cover the night the aerial bombardment started. She lay awake, she says, in a village mosque on the Yemeni island of Birim, counting explosions as the baby cried. It could have been worse though. They could have still been in their house when the first missile landed. ”Our neighbor shouted to my husband ’you have to leave, they’re coming.’ And we just ran. As soon as we left the house, the first missile fell right by it and then a second on it. It burned everything to the ground,” Qassim tells us ...	
GPT-2: Yemen is in the middle of a civil war. Saudi Arabia is leading the coalition bombing campaign. It’s been bombing Yemen for more than two months now.	Reference: Amina Ali Qassim’s family sought shelter in a mosque before fleeing Yemen. Thousands like them are boarding boats to sail to Djibouti. Saudi Arabia has been pounding Yemen in a bid to defeat Houthi rebels.

表14. GPT-2生成的摘要与CNN和Daily Mail数据集测试集上的参考摘要对比。

Language Models are Unsupervised Multitask Learners	
English reference	GPT-2 French translation
This re-release, titled The Next Day Extra, was presented in the form of three disks: the original album, unpublished studio sessions and remixes, plus a DVD containing the four clips that have already been unveiled.	Les nouvelles re-releases, tout en premier disc, nécessaire de l’album, un studio session et remixes, plus une DVD de l’écran de quelques clips qui ont été déjà échappés.
French reference	GPT-2 English translation
Un homme expliquait que le fonctionnement de la hernia fonctionnelle qu’il avait reconnaît avant de faire, le fonctionnement de la hernia fonctionnelle que j’ai réussi, j’ai réussi.	This release, originally released on three discs: the original album, sessions studio inédites and remixes, a DVD containing the four clips that were released on the previous day.
English reference	GPT-2 French translation
One man explained that the free hernia surgery he’d received will allow him to work again.	Un homme expliquait que le fonctionnement de la hernia fonctionnelle qu’il avait reconnaît avant de faire, le fonctionnement de la hernia fonctionnelle que j’ai réussi, j’ai réussi.
French reference	GPT-2 English translation
Un homme a expliqué que l’opération gratuite qu’il avait subie pour soigner une hernie lui permettrait de travailler à nouveau.	A man told me that the operation gratuity he had been promised would not allow him to travel.
English reference	GPT-2 French translation
Speaking to an open government conference in London via video link, Mr Kerry said: ”There is no question that the President and I and others in government have actually learned of some things that had been happening on an automatic pilot because the ability has been there, going back to World War Two and to the very difficult years of the Cold War, and then, of course, 9/11.”	Le premier conference de l’édition internationale en ligne sur la vide, le ministre de l’Édition internationale a été vide, lui dit : ”Il n’y a qu’une question que le président et moi et tous les autres personnes d’une administration qui a été évident par la fois, par la part de la guerre, par la part de la guer
French reference	GPT-2 English translation
S’exprimant lors d’une conférence intergouvernementale à Londres par liaison vidéo, M. Kerry a déclaré: ”Il est indéniable que le Président, moi-même et d’autres membres du gouvernement avons pris connaissance de certaines choses en mode pilote automatique parce que nous en avons la possibilité, dès la Seconde guerre mondiale et jusqu’aux années difficiles de la Guerre froide, puis bien sûr le 11 septembre.”	In a conférence between the United States and London, Secretary of State John Kerry said: ”It is indeniable that the President, myself and others of the government have been aware of certain certain choices that have been made in the past in order to be able to do certain things in a more automated way.”

Table 15. English to French and French to English translations generated by GPT-2.

Language Models are Unsupervised Multitask Learners	
English reference	GPT-2 French translation
This re-release, titled The Next Day Extra, was presented in the form of three disks: the original album, unpublished studio sessions and remixes, plus a DVD containing the four clips that have already been unveiled.	Les nouvelles re-releases, tout en premier disc, nécessaire de l’album, un studio session et remixes, plus une DVD de l’écran de quelques clips qui ont été déjà échappés.
French reference	GPT-2 English translation
Un homme expliquait que le fonctionnement de la hernia fonctionnelle qu’il avait reconnaît avant de faire, le fonctionnement de la hernia fonctionnelle que j’ai réussi, j’ai réussi.	This release, originally released on three discs: the original album, sessions studio inédites and remixes, a DVD containing the four clips that were released on the previous day.
English reference	GPT-2 French translation
One man explained that the free hernia surgery he’d received will allow him to work again.	Un homme expliquait que le fonctionnement de la hernia fonctionnelle qu’il avait reconnaît avant de faire, le fonctionnement de la hernia fonctionnelle que j’ai réussi, j’ai réussi.
French reference	GPT-2 English translation
Un homme a expliqué que l’opération gratuite qu’il avait subie pour soigner une hernie lui permettrait de travailler à nouveau.	A man told me that the operation gratuity he had been promised would not allow him to travel.
English reference	GPT-2 French translation
Speaking to an open government conference in London via video link, Mr Kerry said: ”There is no question that the President and I and others in government have actually learned of some things that had been happening on an automatic pilot because the ability has been there, going back to World War Two and to the very difficult years of the Cold War, and then, of course, 9/11.”	Le premier conference de l’édition internationale en ligne sur la vide, le ministre de l’Édition internationale a été vide, lui dit : ”Il n’y a qu’une question que le président et moi et tous les autres personnes d’une administration qui a été évident par la fois, par la part de la guerre, par la part de la guer
French reference	GPT-2 English translation
S’exprimant lors d’une conférence intergouvernementale à Londres par liaison vidéo, M. Kerry a déclaré: ”Il est indéniable que le Président, moi-même et d’autres membres du gouvernement avons pris connaissance de certaines choses en mode pilote automatique parce que nous en avons la possibilité, dès la Seconde guerre mondiale et jusqu’aux années difficiles de la Guerre froide, puis bien sûr le 11 septembre.”	In a conférence between the United States and London, Secretary of State John Kerry said: ”It is indeniable that the President, myself and others of the government have been aware of certain certain choices that have been made in the past in order to be able to do certain things in a more automated way.”

表15. 由GPT-2生成的英语到法语和法语到英语翻译。

Context (passage and previous question/answer pairs) The 2008 Summer Olympics torch relay was run from March 24 until August 8, 2008, prior to the 2008 Summer Olympics, with the theme of “one world, one dream”. Plans for the relay were announced on April 26, 2007, in Beijing, China. The relay, also called by the organizers as the “Journey of Harmony”, lasted 129 days and carried the torch 137,000 km (85,000 mi) – the longest distance of any Olympic torch relay since the tradition was started ahead of the 1936 Summer Olympics. After being lit at the birthplace of the Olympic Games in Olympia, Greece on March 24, the torch traveled to the Panathinaiko Stadium in Athens, and then to Beijing, arriving on March 31. From Beijing, the torch was following a route passing through six continents. The torch has visited cities along the Silk Road, symbolizing ancient links between China and the rest of the world. The relay also included an ascent with the flame to the top of Mount Everest on the border of Nepal and Tibet, China from the Chinese side, which was closed specially for the event. Q: What was the theme A: “one world, one dream”. Q: What was the length of the race? A: 137,000 km Q: Was it larger than previous ones? A: No Q: Where did the race begin? A: Olympia, Greece Q: Is there anything notable about that place? A: birthplace of Olympic Games Q: Where did they go after? A: Athens Q: How many days was the race? A: seven Q: Did they visit any notable landmarks? A: Panathinaiko Stadium Q: And did they climb any mountains? A: Model answer: Everest Turker answers: unknown, yes, Yes, yes
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Table 16. Selected CoQA completion.

上下文（段落及先前的问题/答案对） 2008年夏季奥运会火炬接力于2008年3月24日至8月8日举行，主题为“同一个世界，同一个梦想”。接力计划于2007年4月26日在中国北京公布。此次接力被主办方称为“和谐之旅”，历时129天，火炬传递里程达137,000公里（85,000英里）——这是自1936年夏季奥运会开创火炬接力传统以来，距离最长的一次奥运火炬传递。 3月24日在希腊奥林匹亚的奥运会发源地被点燃后，火炬传递至雅典的帕纳辛奈科体育场，随后于3月31日抵达北京。从北京出发，火炬沿着一条穿越六大洲的路线行进。火炬途经丝绸之路沿线城市，象征着中国与世界之间古老的联系。此次接力还包括从中国一侧，专门为此活动封闭的尼泊尔与中国西藏边境的珠穆朗玛峰顶峰进行火焰登顶。 问：主题是什么？ 答：“同一个世界，同一个梦想”。 问：比赛的长度是多少？ 答137,000公里 问：比之前的更大吗？ 答：不是 问：比赛从哪里开始？ 答：希腊奥林匹亚 问：那个地方有什么特别之处吗？ 答：奥运会的发源地 问：之后他们去了哪里？ 答：雅典 问：比赛进行了多少天？ 答：七天 问：他们参观了哪些著名地标吗？ 答：帕纳辛奈科体育场 问：他们爬过山吗？ 答：参考答案：珠穆朗玛峰 众包回答：未知，是的，是的，是的

表16. 选定的CoQA完成情况。

Context (passage and previous question/answer pairs) Tom goes everywhere with Catherine Green, a 54-year-old secretary. He moves around her office at work and goes shopping with her. ”Most people don’t seem to mind Tom,” says Catherine, who thinks he is wonderful. ”He’s my fourth child,” she says. She may think of him and treat him that way as her son. He moves around buying his food, paying his health bills and his taxes, but in fact Tom is a dog. Catherine and Tom live in Sweden, a country where everyone is expected to lead an orderly life according to rules laid down by the government, which also provides a high level of care for its people. This level of care costs money. People in Sweden pay taxes on everything, so aren’t surprised to find that owning a dog means more taxes. Some people are paying as much as 500 Swedish kronor in taxes a year for the right to keep their dog, which is spent by the government on dog hospitals and sometimes medical treatment for a dog that falls ill. However, most such treatment is expensive, so owners often decide to offer health and even life _ for their dog. In Sweden dog owners must pay for any damage their dog does. A Swedish Kennel Club official explains what this means: if your dog runs out on the road and gets hit by a passing car, you, as the owner, have to pay for any damage done to the car, even if your dog has been killed in the accident. Q: How old is Catherine? A: 54 Q: where does she live? A: Model answer: Stockholm Turker answers: Sweden, Sweden, in Sweden, Sweden

Table 17. Selected CoQA completion.

上下文（段落及先前的问题/答案对） 汤姆与54岁的秘书凯瑟琳·格林形影不离。他在她的办公室走动，陪她购物。凯瑟琳觉得汤姆很棒，她说：“大多数人似乎都不介意汤姆。”她说：“他是我的第四个孩子。”她可能把他当作自己的儿子来对待。他四处走动，购买食物，支付医疗账单和税款，但事实上汤姆是一只狗。 凯瑟琳和汤姆住在瑞典，这个国家期望每个人都按照政府制定的规则过着井然有序的生活，政府也为人民提供了高水平的关怀。这种关怀水平需要金钱来支撑。 瑞典人对一切事物都要交税，所以养狗意味着要交更多的税，他们对此并不感到惊讶。有些人每年要支付高达500瑞典克朗的税来获得养狗的权利，这些钱由政府用于狗医院，有时也用于治疗生病的狗。然而，大多数此类治疗费用昂贵，因此狗主人常常决定为他们的狗提供健康乃至生命的保障。 在瑞典，狗主人必须为他们的狗造成的任何损害支付赔偿。瑞典养犬俱乐部的一位官员解释了这意味着什么：如果你的狗跑到马路上被过往的车辆撞到，作为主人，你必须赔偿对车辆造成的任何损害，即使你的狗在事故中已经死亡。 问：凯瑟琳多大了？ 答： ： 54岁 问：她住在哪里？ 答： ： 模型答案：斯德哥尔摩 土耳其人答案：瑞典，瑞典， ，在瑞典，瑞典
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Table 17. Selected CoQA completion.